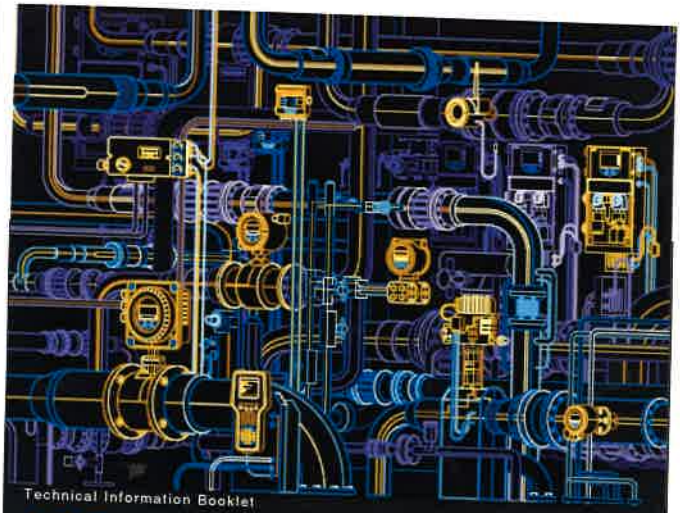


Technical Information Booklet

ABB Instrumentation
Flowmeter Handbook

Power and productivity
for a better world™

ABB

ABB FLOW SOLUTIONS

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ABB Flowmeter Sizing Software (Genie II) 2

Online Product Information Library

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CD-CALC SIZING SOFTWARE

ABB offers CD-Calc sizing software for TrioWirl Vortex and Swirl, Variable Area, Magnetic, and TrioMass coriolis meters. CD-Calc allows you to quickly select the correct size meter for your application. It is easy to use, has a built-in HELP function, and provides a great deal of useful information in addition to the correct size meter. You can obtain a free download CD-Calc using the following procedure:

- go to www.abbinfozone.com
- enter infozone from the main page
- select 1cFL-Flow - Vortex & Swirl from the Product Group pick list
- select "software downloads" from the Type of Information pick list
- click Index and follow the instructions on the screen

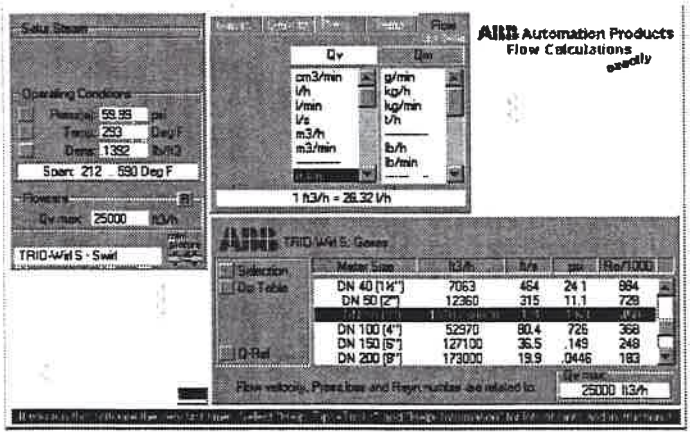


ABB FLOWMETER SIZING PROGRAM (GENIE II)

ABB offers the Genie II sizing software for wedge elements, integral orifice elements, orifice plates, venturi elements, flow nozzles, pitot tubes and elbow elements. Genie II allows you to quickly select the size element (and restriction) to create a given differential pressure for any application. Genie II is easy to use, has a built-in tutorial function, and many useful features, including fluid physical properties. You can obtain a free download of Genie II using the following procedure:

- go to www.abbinfozone.com
- enter infozone from the main page
- on the left of the next screen you will see the heading "Other Usefull ABB Instrumentation Sites." Click on the item "ABB Flowmeter Sizing Program"
- follow the instructions on the screen

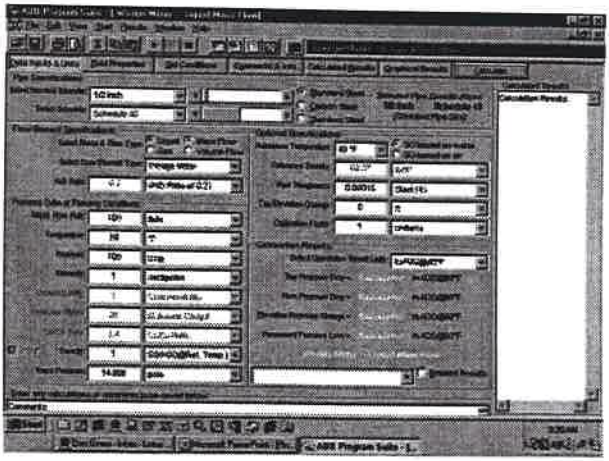
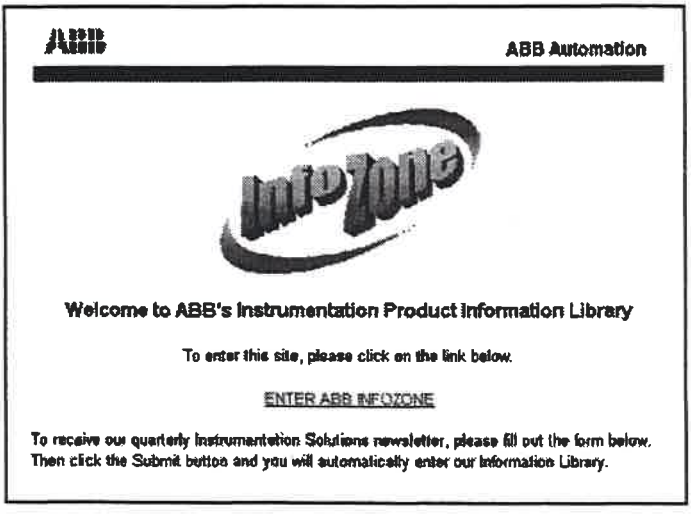


ABB INFOZONE

Information about any ABB products may be found at www.abbinfozone.com. The following is available for free download:

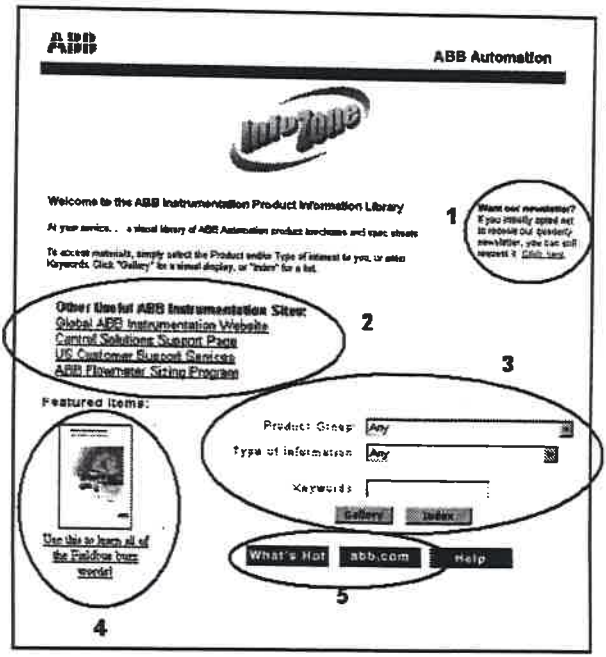
Products
Flow Products
Pressure Transmitters
Temperature Instruments
Recording & Control Products
Control Solutions
Analytical Products
Valve Automation Instruments
Level Instruments

Type of Information
Ads
Brochures
Case histories
Instruction Manuals
Newsletters
QA Procedures
Software downloads
Specification Sheets
Technical Information



Online Product Information Library

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2. Quick links to other useful ABB sites
3. Use the Product Group and Type of Information pick lists and then click Gallery or Index for search results
4. Click here to see Featured items
5. Quick links to "What's Hot!" and abb.com

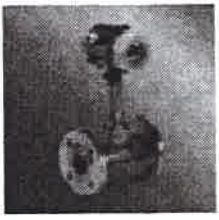


TrioWirl Vortex Shedding and Swirl Flowmeters with Digital-Signal-Processing

- No Moving Parts
- Immunity to vibration and noise
- Integral temperature measurement
- Common sensor and electronics for all size vortex and swirl meters
- Digital communications with HART, FOUNDATION FIELDBUS or PROFIBUS options



TrioWirl Series 4000 vortex shedding flowmeters are the most cost effective solution for many liquid, gas and steam applications. They are available with wafer or flanged style bodies in sizes 1/2 to 12 inch.



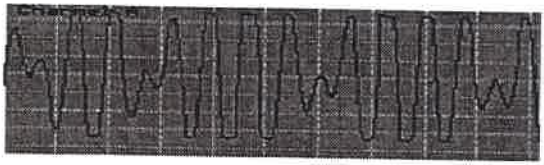
TrioWirl Series 4000 swirl flowmeters are preferred to vortex meters when the application calls for lower flow measurement capability, for higher accuracy (0.5%) on gases and steam, and for high accuracy on tight installations offering minimal straight runs of pipe upstream of the meter. Swirl meters are available in flanged style bodies in sizes from 1/2 to 16 in.

TrioWirl features common to both vortex and swirl meters include an option for an integral temperature measurement, stainless steel or Hastelloy C construction, meter bodies fully rated for ANSI Class 600 flanges, remote mounted electronics, and compatibility with process temperatures up to 536° F.

DSP - A Giant Leap In Performance

Pipeline vibration and hydraulic noise are always present to some degree in any process and can make it difficult to measure accurately with conventional vortex meters. TrioWirl vortex and swirl meters with ABB's state-of-the-art DSP technology provides accurate measurement immune from the effects of vibration and noise.

The graph below is a typical vortex sensor signal from a process with noise and/or vibration. The collapsing peaks of the sine wave signal can cause inaccurate measurement in other meters.



However, the graph below shows the stability of the TrioWirl output from the same "noisy" signal after digital signal processing. You don't have to live with "jittery" and inaccurate output, limited turndown, or wieldy pipe supports.

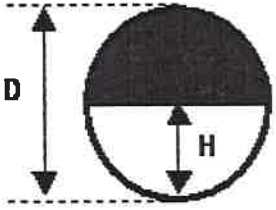


Find information on TrioWirl Vortex Shedding and Swirl meters at www.abbinfozone.com.

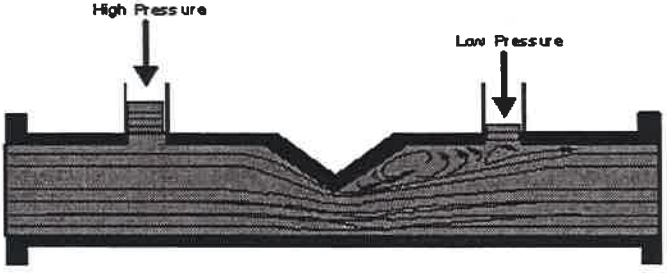
Wedge Primary Flow Elements

A Wedge element has an internal "v" shaped restriction to flow. This restriction produces a differential pressure which follows Bernoulli's square root relation to flowrate over Reynolds numbers greater than 500. Wedge elements offer higher accuracy over wider flow ranges (laminar, transition, or turbulent) than any other primary element. It can handily measure viscous liquids or liquids with varying viscosity.

Wedge elements are available in sizes from 1/2 in. to 24 in. Larger sizes are available upon request. The Wedge element is classified by the H/D ratio, analogous to the Beta ratio for an orifice plate. The H/D ratio is the height of the open area available for flow divided by the inside pipe diameter.



Wedge elements offer a calibrated accuracy of 0.5%. The Wedge restriction has no critical surface dimensions, or sharp edges that must remain within strict tolerances for maximum measurement accuracy.



This restriction is designed to last the life of the installation, without replacement and without recalibration. Wedge systems with the ABB Multivariable DP transmitter can measure flow ranges of 3.5: 1 (12:1 in DP) with an overall system accuracy of under 1% of rate, or flow ranges of 5:1 (25:1 in DP) with an overall system accuracy of under 2% of rate. "Stacking" DP transmitters (with different ranges) on the same Wedge element can provide even wider flow ranges and/or higher accuracy.

Wedge elements are available in a wide assortment of process connections to easily install in any piping system. In addition, the Wedge element is highly insensitive to velocity profile distortion and swirl and requires minimal straight runs of upstream/downstream piping. In addition to wedge element with standard pipe taps for clean fluids, wedge elements are also compatible with DP transmitters having remote seals. A remote seal Wedge element is used for tougher applications, like slurries (and the wastewater application cited earlier), where it is not feasible from a maintenance viewpoint to use impulse tubing for the DP transmitter connection. The Wedge element and remote seal transmitter combination keeps the process fluid contained within the pipe and offers no dead zones that can lead to clogging.



Find information on Wedge elements at www.abbinfozone.com.

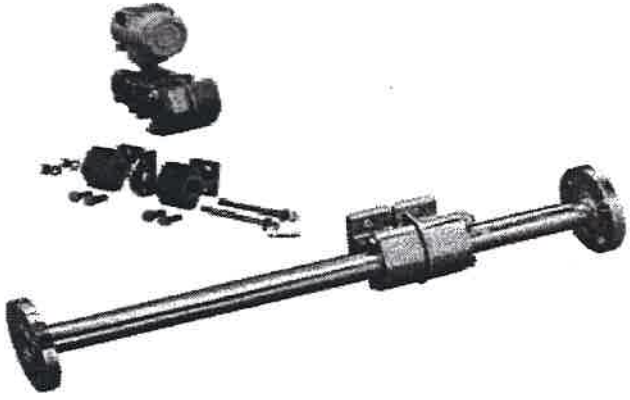
Integral Wedge element (left), pipe tap Wedge element (center), remote seal Wedge element (right)

Integral Orifice Elements

The 1/2 in., 1 in., and 1.5 in. integral orifice elements meter low flows of clean liquids, gases and steam. The orifice plate is a special design that clamps between two meter body halves. The 1/2 and 1 in. integral orifice is rated to a maximum of 3000 psi, and the 1.5 in. is rated to 1500 psi. The integral orifice meter body connects directly to a differential pressure transmitter or to a 3-valve manifold (which in turn mounts to the transmitter). This greatly simplified installation over a conventional paddle type orifice plate by eliminating the need for any pipe fittings or impulse tubing. The installation is much easier to maintain and service. The integral orifice is also available as a flow element with attached straight runs of upstream and downstream piping.

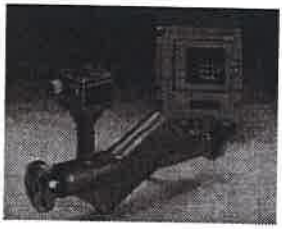
Integral orifice elements offer calibrated accuracy of 0.5%, a wide variety of standard bore options, and 316 stainless steel meter bodies (stainless steel or Hastelloy C plates).

Find information on integral orifice elements at www.abbinfozone.com

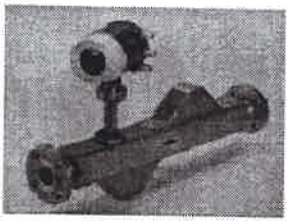


TRIOMASS CORIOLIS MASS FLOWMETERS

TrioMass coriolis mass flowmeters from ABB are cost effective solutions to liquid mass flow and/or density measurement. Sensors are offered in sizes from 3/8 in. To 6 in. and incorporate a compact, parallel-path, dual-tube design. The rugged flowtubes provide secondary containment as a standard feature. Stainless steel and Hastelloy C construction are offered. The flowtubes are self-draining and suitable for clean-in-place sanitary applications.



The MC1 flowtubes are used with remote mounted 50MM2000 flow computers. The 50MM2000 is full-featured, has a large display with integral keypad, and user friendly menu driven software. Continuous flow, batch operation and density software options are available.



MC2 flowtubes are used wit the new DSP converters. The MC2 flowtube has the same design as the MC1 but is integrally connected to the DSP converter. The DSP TrioMass reduces installation time, will have options for higher precision for mass rate and density, and explosion proof converter housing.

TrioMass systems are available in a wide range of sizes and flow ranges. They offer high reliability and precision and excellent long term stability

Find more information on TrioMass flowmeters on www.abbinfozone.com

ABB Industrial Magnetic Flow Meters

ABB offers a complete line of electromagnetic flowmeters that provide cost effective solutions to virtually any application involving a conductive liquid or slurry.

10D1475 wafer sensor and 10DX4311 flanged sensors with integrally mounted XE transmitters are the most cost effective system for clean liquids.



10DT43 flanged sensors with integral XT transmitters are two-wire, loop powered systems for clean liquids that simplify and reduce the cost of installation.



10D1475 wafer sensors and 10D3111 flanged sensors with remote mounted 50XM transmitters are full featured systems recommended for clean liquids, dirty liquids, and low noise producing slurries.



10D1476 wafer sensors with ceramic liners and remote mounted 50XM transmitters are recommended on clean liquids, highly corrosive liquids, dirty liquids, abrasive fluids, and low noise producing slurries. There is a 1/25 in. sensor that is useful for measuring very low flowrates.



10D1477 electrodeless sensors with ceramic liners and remote mounted 50XM transmitters are recommended for low conductivity fluids (less than 5 microsiemens/cm), low noise producing slurries, noisy slurries, and fluids that product coatings.

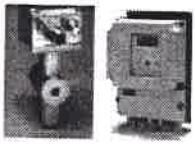
The 10DX3121 flanged electrodeless Tefzel lined sensor and remote mounted 50CM transmitter are recommended for low conductivity fluids (less than 5 microsiemens/cm), low noise producing slurries, noisy slurries, and fluids that can produce coatings.



10DS3111 flanged sensor and remote mounted 50SM transmitters are AC magnetic flow systems. They are recommended for all slurries, particularly applications involving pulsating flow and fast speed of response.



DM21 wafer sensors and remote mounted 50XM transmitters are recommended for sanitary applications. These 3A sensors have stainless steel exteriors and are available with Tri-clamp quick disconnect fittings. A Pharmaceutical design option is available that features a continuous bore and crevice free adaptors. The DM21/50SM is an AC magnetic flow system with the same features as the DM21/50XM.



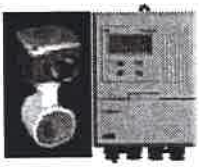
Find information about these and other ABB magnetic flowmeters at www.abbinfozone.com. You may also obtain a free multi-media interactive CD entitled " ABB Magnetic Flowmeter Selection Guide" which has complete product and technical information on all ABB magnetic flow products and which helps you to select the correct flow system for any specific application.

ABB Municipal Magnetic Flow Meters

ABB offers the MagMaster and AquaProbe lines of electromagnetic flowmeters to provide cost effective solutions to virtually any application in the Municipal industries involving a conductive liquid or slurry.

MagMaster pulsed DC magnetic flow systems include 10D1475 series wafer sensors, and MFE/MFF series flanged sensors combined with the MagMaster MFE transmitter. MagMaster systems offer a variety of materials of construction and have the options and features to meet the requirements of virtually all applications. This includes buriable and submersible sensors, bi-directional flow capability, and serial communications, just to name a few.

10D1475 wafer sensors and remote mounted MagMaster transmitters are recommended for water and waste water applications except high noise producing slurries or sludges and other fluids that produce coatings. This system is supplied for general purpose service.



MFE flanged sensors and MagMaster transmitters are recommended for all water and waste water applications. These systems can be supplied for general purpose or hazardous area service

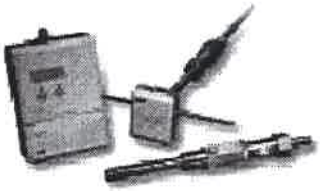


MFF flanged sensors and MagMaster transmitters are recommended for all water and waste water applications. These systems are supplied for general purpose service



AquaProbe insertion magnetic flowmeters are ABB's most cost effective solution for flow measurement in pipes 8 in. and larger. AquaProbe sensors are recommended when the process can not be shut-down for installation, for installation in buried pipe with minimal excavation, for use as a portable meter, or to perform in-line verification of other flow devices. AquaProbes require longer runs of straight piping upstream of the meter (than do MagMaster systems) to ensure good hydraulics and maximum performance. AquaProbes are recommended for clean liquids and for general purpose service only.

AquaProbe I is supplied with a MagMaster transmitter and is recommended when commercial power is readily available.





AquaProbe II is supplied with an AquaMaster transmitter and is recommended when battery powered instrumentation is needed.

Find information about these and other ABB magnetic flowmeters at www.abbinfozone.com. You may also obtain a free multi-media interactive CD entitled "ABB Magnetic Flowmeter Selection Guide" which has complete product and technical information on all ABB magnetic flow products and which helps you to select the correct flow system for any specific application.

Rotameters

ABB offers an extensive range of rotameters, each providing the durable construction, reliability and accuracy demanded by your process. Variable area flowmeters are designed to measure the flow of liquids or gases via a closed pipe system and a float.

The series 10A3200 Amored Purgemeter is an all-metal stainless steel flowmeter specifically designed for low-flow measurement of most liquids and gases at high pressure and high temperatures. It can be applied to a wide range of applications in the Municipal, Chemical, Petrochemical, Pharmaceutical and process industries. This easy to install meter offers:

- One wetted moving part
- Compatibility with dirty and corrosive gases and liquids
- Measurement ranges from 0.3 to 211 GPH on water; 1.3 to 880 SCFH on air
- Minimum and/or maximum alarm options
- Analog output
- Integral 1/4 in. needle valve
- Differential pressure regulator
- 1/4, 3/8 and 1/2 in. NPT process connection options



The series 10A4500 VA Master Flowrator is a glass-tube flowmeter. The 10A4500 meters are ideal for measuring a wide range of liquids and gases found n most industrial facilities. Applications include monitoring natural gas flows into ovens and furnaces, cooling fluid to protect equipment, sampling systems in laboratories, and flowrates into and out of large tanks. It has a stainless steel body and these features:

- Universal threaded process connection
- Easy removal of glass tube
- Easy range change and cleaning
- 1/2 to 2 in. bore tube sizes for liquids and gases
- Visual indication of flow over 12.5:1 range (linear scale)
- Screw-on flanges
- Rotatable end fittings
- O-ring or packing gland seals
- Optional one or two b—stable alarms for contact closure on rising or falling flow



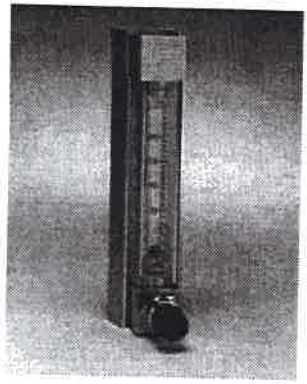
Rotameters (continued)



The series AM54 Armored VA Meter is a flow-through meter designed to measure liquids, steam and gas. The meter is offered in stainless steel or Hastelloy C and is suitable for a wide variety of applications. It is recommended for high pressure and high temperature environments, and is ideal for the chemical, pharmaceutical and food industries.

The AM54's float is magnetically coupled to an indicator alone or with electronic output transmission or an electronic alarm. 4-20 mA ouput, HART communications and a scaled pulse output are available. This magnetic coupling uses span, zero and linearity adjustments enable you to use linear scales for percent of full-scale flow or direct reading units. Key features include:

- Assorted metering tubes and floats
- 1/2 in. to 4 in. diameters
- Measurement ranges from 0.1 to 530 GPH liquid; 0.5 to 2200 SCFM gas
- Wide array of flange connections
- Straight-through design
- Short face-to-face dimensions
- Easy-to-replace, heavy gauge float stop
- Optional Teflon lining and Teflon float
- Spiral upper guide rods on gas service



The series 10A6100 Purgemaster is a low capacity meter for both liquid and gas. The Purgemaster is ideal for many low flow rate applications, such as purging control lines and instrument enclosures. The Purgemaster is also excellent for fluid sampling, and liquid specific gravity and level measurements.It has a stainless steel body and offers the followng features:

Simple snap-in tube construction

- Four scale lengths
- Interchangable parts for all four lengths
- Internal backcheck to restrict backflow and draining when tube is removed
- Optional flow controller
- Optional one or two bi-stable alarms for contact closure or opening on rising or falling flow

Find information on Rotameters at www.abbinfozone.com including the 12-page brochure "Rotameter Selection Guide".

SELECTION AND APPLICATION

Primary Flow Element Selection

You should determine the answer to these five questions when selecting an element:

1. The Measured Fluid

- Liquid, gas, or vapor (steam)?
- One phase or two phase flow?
- Clean, dirty, a slurry, viscous, corrosive, or erosive?

2. The Process Conditions

- Operating temperature and pressure?
- Upset conditions?
- Constant or varying conditions?
- Reynolds number range (flowrate, density, viscosity)?

3. The Installation Requirements

- Line size?
- Upstream and downstream piping?
- Vibration and/or pulsating flow?
- Ambient conditions?

4. The Performance Required

- Accuracy?
- Flow range?
- Speed of response?

5. Economic Considerations

- Initial cost (onetime)?
- Installation cost (onetime)?
- Maintenance cost (annual)?
- Operating cost (annual)?

Normally there is an interaction of some or all the above factors. Final selection will be a compromise between desired features and equipment cost.

Application and Performance Comparison

Table 1 compares those flowmeters most commonly used for typical industrial applications. This table allows you to quickly select which meters are best suited to your particular application. The information in this guide represents generally accepted concepts of operation for each meter on a generic basis. information distributed by specific manufacturers may vary slightly.

Table 1. Flowmeter Applications and Features

Key: ● Recommended ◉ Limited Applicability ○ Not Recommended N - None L - Low M - Medium H - High	Applications								Features									
	Clean Liquid	Dirty Liquid	Viscous Liquid	Corrosive	Slurries	Clean Gas	Dirty Gas	Steam	Sizes Available	Optimal Accuracy	Nominal Rangeability	Reynolds Numbers Limitations	Pressure Loss	Relative Cost	Nominal Upstream Piping Requirements	Installation	Maintenance	Type of Output
Orifice Plate	●					●		●	>1"	3/4%	3/1	>30,000	H	L	10-30D	M-H	M-H	√
Integral Orifice	●					●		●	1", 1.5"	2%	3/1	>10,000	H	L	10-30D	L	M-H	√
Wedge™	●	●	●	○	●	●	●	●	>0.5"	1/2%	3/1	>500	M	M	5-10D	L	L	√
Flow Nozzle	●	○		○		●	○	○	>2"	1-1/2%	3/1	>75,000	H	M	10-30D	M	L	√
Venturi Tube	●	○		○	○	●	○	○	>2"	1%	3/1	>75,000	L	H	5-10D	M	L	√
Flobar™	●	○		○		●	○	●	>0.5"	1%	3/1	>40,000	L	L	10-30D	L	L	√
Magnetic	●	●	●	●	●				>1"	1/2%	10/1	None	L	H	5D	H	M	Linear

Application and Performance Comparisons

Rotameter	●	○	○	○		●		○	≤3"	2%	10/1	None	M	L	None	L	L	Linear
Turbine	●		○	○		●		●	>0.25"	1/2%	10/1–50/1	≤2–15 cSt	H	M	10–20D	L	M-H	Linear
Positive Displacement	●	○	●	○		●			<12"	1%	20/1	≤8000 cSt	H	H	None	H	H	Linear
Vortex Shedding	●	○		○		●	○	●	>1"	1%	20/1	>10,000	M	M	15–25D	L-M	M	Linear
Doppler		○	●	○	●				>0.5"	2-5%	1-/1	None	L	M	5–20D	L	L	Linear
Transit Time	●		○	●					>0.5"	2-5%	10/1	None	L	M	5–20D	L	L	Linear
Mass – Coriolus	●	●	●	○	●	○	○		<6"	1/4%	25/1	None	M	H	None	H	L-M	Linear
Target	●	●	●	○		●	●		>0.5"-4"	1-1/2-5%	3/1	>100	M	L	10–20D	L	M	√

Differential Element Flow Equations

The equations used to calculate the differential pressure created by an orifice plate, integral orifice assembly and wedge element are listed below. Direct calculations can be made using these equations.

Orifice Plate, Integral Orifice Assembly

$$h = g_f \left(\frac{q\,(U.S.\ gpm)}{5.668 \cdot F_a \cdot K \cdot d^2} \right)^2 \quad \textit{liquid}$$

$$h = \frac{G \cdot T_f}{P_f} \left(\frac{Q\,(scfh)}{7727 \cdot F_a \cdot F_{pv} \cdot Y \cdot K \cdot d^2} \right)^2 \quad \textit{gas}$$

$$h = \overline{V} \left(\frac{W\,(lb/hr)}{359 \cdot F_a \cdot Y \cdot K \cdot d^2} \right)^2 \quad \textit{steam}$$

Wedge Element

$$h = g_f \left(\frac{q\,(U.S.\ gpm)}{5.668 \cdot F_a \cdot Kd^2} \right)^2 \qquad liquid$$

$$h = \frac{G \cdot T_f}{P_f} \left(\frac{Q\,(scfh)}{7727 \cdot F_a \cdot F_{pv} \cdot Y \cdot Kd^2} \right)^2 \qquad gas$$

$$h = \overline{V} \left(\frac{W\,(lb/hr)}{359 \cdot F_a \cdot Y \cdot Kd^2} \right)^2 \qquad steam$$

Table 30. ASME Piping Recommendations – All Fittings in Same Plane

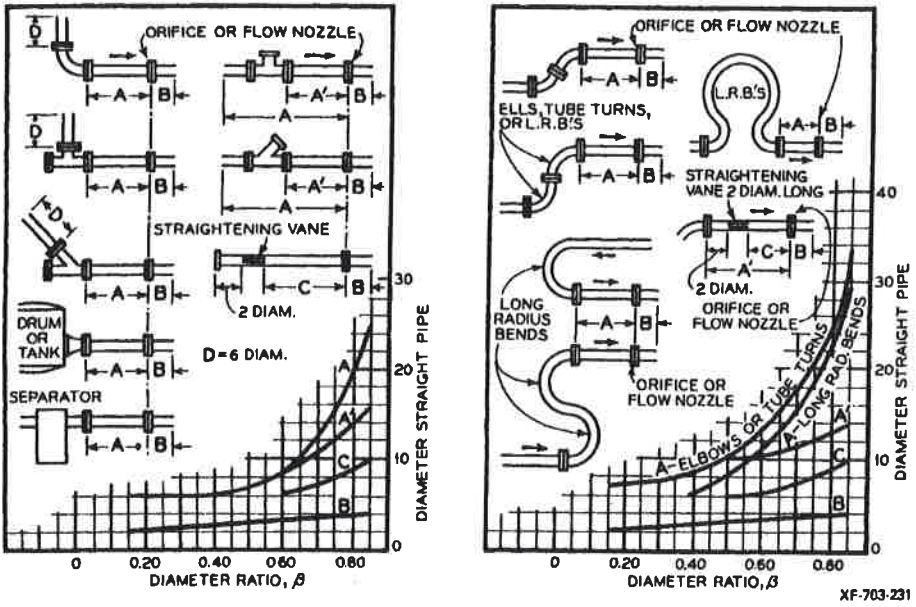
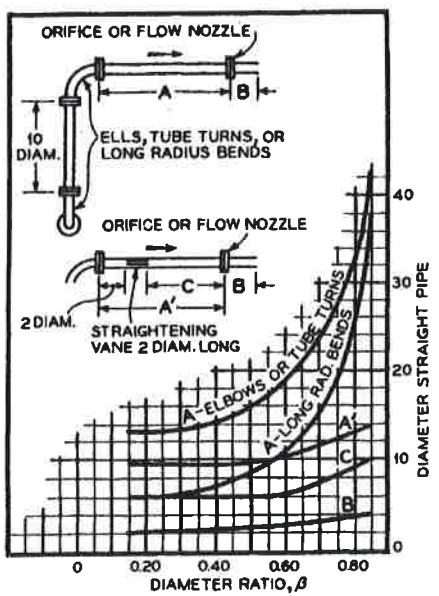


Table 31. ASME Piping Recommendations – Fittings in Different Planes



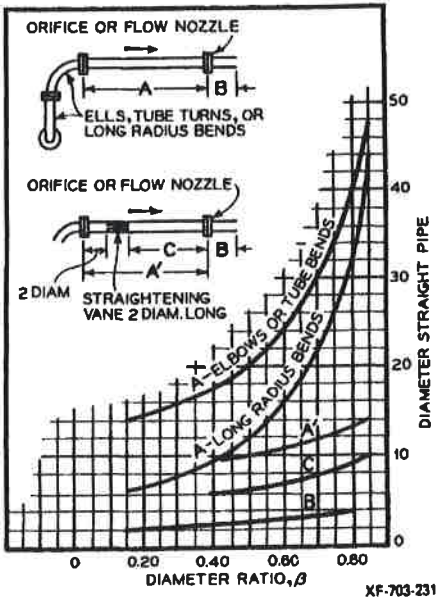


Table 32. ASME Piping Recommendations – With Reducers & Expanders

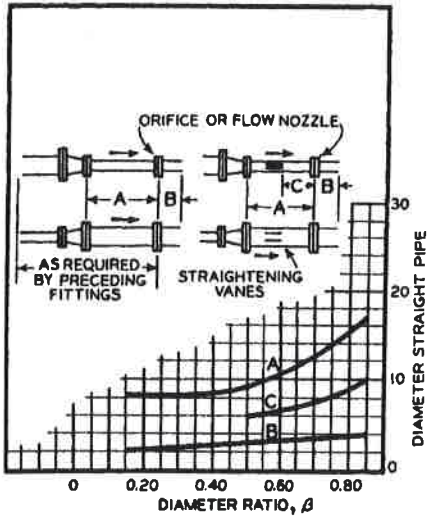


Table 33. ASME Piping Recommendations – Atmospheric Intake

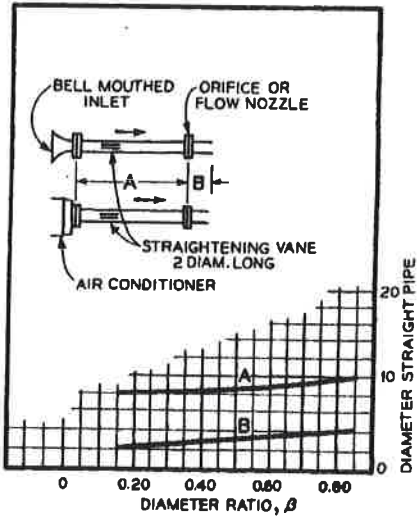
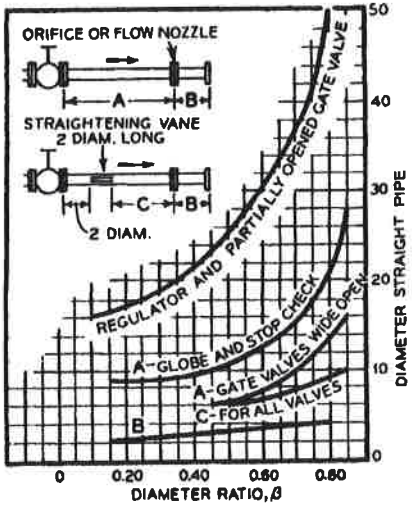
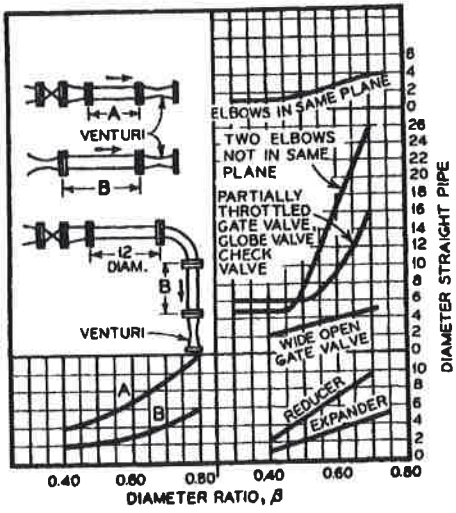


Table 34. ASME Piping Recommendations – Valves & Regulators



XF-703-231

Table 35. ASME Piping Recommendations – Venturi Tubes



XF-703-231

Definition of Terms

Accuracy - A statement which defines the measurement error limit when a device is used under specific operating conditions. It is usually measured as inaccuracy and expressed as accuracy. Flowmeters can be specified with percent of rate or percent of full scale accuracy. Accuracy expressed as a percent of rate implies accuracy anywhere within the flow range of the device. Accuracy expressed as a percent of span applies only at the **maximum** rate of flow of the device. The absolute error obtained at full span is used to calculate accuracy at lower flow rates.

Consider two flowmeters reading between 0 and 100 GPM. One meter has a ± 1 % of rate accuracy, and the other has a ± 1 % of span accuracy. At max flow the meter accuracies are identical. However at 10 GPM, the percent of rate device accuracy is 10 GPM ± 0.1 GPM while the percent of span device accuracy is 10 GPM ± 1 GPM.

Beta Ratio - Represents the ratio of the meter bore diameter (d) to the pipe bore diameter (D). A meter with a bore of 2.013" installed in a 4" Schedule 40 pipe (I.D. = 4.026) would have a Beta Ratio of 0.5. Low Beta ratios relax tolerances on the configuration and installation of the primary elements, but raise the pressure loss through the restriction and reduce meter capacity.

Density - The property of a fluid expressed as weight or mass per unit volume. Density is a function of temperature and pressure. For some primary elements (differential producers), density is part of the flow equation and directly affects measured rate. As an example, water at 60' F has a density of 62.34 pounds per cubic foot. At 200' F, the density is 60.13 pounds per cubic foot.

Differential Pressure -The difference in absolute pressure between the upstream and downstream sides of a restriction to flow. Differential pressure is normally measured in inches of water.

Discharge Coefficient - No primary element follows Bernoulli's Equation or Pouiseville's Law precisely. In an ideal world, void of friction and physical phenomena, the Theory of Continuation would say thatthe flow rate at any point in the piping system isexactlythe same regardless of the pipe size. In the real world, we must take these effects into consideration and account for them. Each and every meter has it's own factor that accounts for the real world. This factor is called the Discharge Coefficient. It is a ratio of the ACTUAL RATE OF FLOW to the THEORETICAL RATE OF FLOW and is determined through laboratory calibration, usually by the weight-time method.

Flow Coefficient - The flow coefficient, K, is an efficiency term defined as K = C/(l-B4)112. This is a single parameter which accounts forthe discharge coefficient and Beta Ratio.

Fluid - Fluids are substances that have the capability to flow and conform to the shape of their containing vessels. All fluids have some degree of compressibility and offer little resistance to change of form. Fluids are normally categorized as liquids, gases or vapors (i.e., steam). Liquids are practically incompressible and occupy definite volumes. Gases and vapors are compressible and fill any volume in which they are placed.

Head Meter - A meter where the fluid stream creates a differential pressure across the primary element due, mainly, to the velocity and density of the fluid.

Head Loss - Every meter uses energy in the form of pressure, or head loss, to the system. In addition to primary element, all fittings, valves, and even straight piping, reflect some degree of head loss. All sources of head loss must be accounted for, and an equivalent amount of energy imposed on the fluid (by pumps and compressors) to get flow through the system.

Laminar Flow - Values of Reynolds Number below 2100 characterize laminar flow. The velocity profile for fully developed laminar flow is parabolic, with maximum fluid velocity occurring at the centerline of the pipe, and decreasing to zero at the pipe wall. The fluid velocity at centerline is twice the value of the average velocity in the pipe.

Linear Meter - A primary element which has an output that is directly proportional to flow rate. Magnetic flowmeters, turbine meters, vortex meters, etc., are all examples of linear devices. By contrast, an orifice plate is a non-linear device, with flow rate being a square-root function of the differential pressure.

Mass Flowimeter - Measures the true weight of a fluid per unit time directly and without any compensation for fluid properties. An example would be a Coriolis Force device.

Master Meter Calibration - Compares the output of the device in question against a standard. The output of the meter being calibrated is adjusted until it agrees with the standard.

Newtonian Liquid - A liquid which has a constant viscosity at a given temperature regardless ofthe rate of shear.

Open Channel Flow - Gravity flow through pipes, weirs, or flumes where the liquid stream is not completely enclosed by solid boundaries.

Pipe Runs - Lengths of unobstructed, straight pipe with smooth internal surface finish located upstream and downstream of a primary element. The object of a pipe run is to eliminate velocity profile distortions and swirl. The particular fittings used in the system and the sensitivity of the primary element to velocity profile distortion determine the length of runs required for proper performance.

Pressure -The force created by fluids on a surface is called pressure. The unit of measure is force-per-unit-area. The way pressure is observed, or the way it is discussed, can vary depending on specific needs.

As examples:

1. The weight of a one square inch column of air above the earth's surface is called ATMOSPHERIC PRESSURE. While 14.7 Pounds per Square inch is generally used for this measurement, it will depend upon the elevation of the area in which the measurement is taking place.

2. GAUGE PRESSURE is a force-per-unit-area that is above or below Atmospheric Pressure. Gauge pressure is always referenced to Atmospheric Pressure and is a measure of deviation from Atmospheric Pressure.
3. ABSOLUTE PRESSURE is a TOTAL PRESSURE and is the amount of pressure over and above a full vacuum (- 14.7 psi). This measurement is not referenced to Atmospheric Pressure, which changes daily.
4. VACUUM is a measure that indicates pressure **below** Atmospheric Pressure. The amount of vacuum indicates how far below. For example 10 inches of Mercury Vacuum is a **higher pressure** than 15 inches of Mercury Vacuum, but a **lower vacuum.**

Pressure Recovery - The pressure decrease as fluid passes through a restriction is a result of a conversion of potential energy to kinetic energy. Some of this kinetic energy is dissipated in turbulence and by friction. The rest is converted back to potential energy downstream of the restriction, so a portion of the pressure drop is recovered. Pressure recovery is determined by the geometry of the primary element.

Prover Calibration - Generally provided on signal generators (Magnetic, Turbine, Vortex, etc.). A known volume is passed through the meter in question and the meter's output compared to the known volume. The volume is usually provided by a ballistic prover, which is essentially a piston and cylinder. A precisely bored cylinder is evacuated by a piston with a precise stroke length.

Primary Element - A device, usually in the form of a restriction, inserted in the pipe line to develop a mechanical, electrical, or hydraulic force for purposes of measuring flow.

Pulsating Flow - Flow is considered steady if cyclic changes in characteristic variables are slower than approximately one-fifth of the undamped natural frequency of the measuring system. Pulsating flow is usually caused by reciprocating or rotating pumps, compressors, turbines, chattering valves or liquid surges in low portions of gas or

vapor lines. Flowmeters usually provide indications greater than actual flows under pulsating flow, and a meter with a rapid response time is required to minimize error

Rangeability - The ratio of maximum flow to minimum flow over which the specified accuracy of a primary element will be maintained.

Repeatability - The ability of a meter to indicate the same reading each timethe same flow condition exists.

Resolution - A measure of the smallest increment of total flow that can be measured.

Reynolds Number - The dynamic behavior of fluids depends heavily on the relative magnitudes of frictional and inertial forces. Frictional forces are dominant in low velocity streams and tend to produce laminarflow. Inertial forces are greater when velocities are high and produce turbulent behavior. The Reynolds Number is a single dimensionless parameter formed as the ratio of inertial to viscous forces. The magnitude of this term indicates whether flow is laminar orturbulent and the subsequent shape of the velocity profile.

The Reynolds Number formulae for liquid, gas and steam follow:

$$RD = \frac{(3160)(q)(gf)}{(D)(u)} \quad \text{Liquid}$$

$$RD = \frac{(0.482)(Q)(G)}{(D)(u)} \quad \text{Gas}$$

$$RD = \frac{(6.32)(W)}{(D)(u)} \quad \text{Steam}$$

Secondary Element - A transducer wh ich a m plif ies the signa l f rom a primary element. This unit may be mechanical, electrical, or pneumatic.

Swirl - Represents angular momentum of the fluid that results in rotational flow about the pipe axis. Swirl is the effect of fluid flow through different combinations of pipe fittings, such as two close coupled 90' elbowsout of plane.

Transition Flow - Characterized by Reynolds Numbers between 2100 and 3000. Flow takes on the properties of both laminar and turbulent characteristics in this region.

Turbulent Flow - Values of Reynolds numbers from 3000 and higher characterize turbulent flow. As Reynolds Number increases from the laminar to turbulent region, the velocity profile flattens. Eventually, all fluid particles travel at a uniform velocity (except near the pipe walls). This is also called plug flow.

Velocity Profile Distortion -Any profile (either turbulent or laminar) that is not symmetric about the center axis of the pipe. The result is a slanted profile with fluid flowing faster along one side of the pipe than along the other side. Such distortions are caused by flow through fittings, partially closed valves, gaskets extending into the f l ow, etc.

Vena Contracta - The vena contracta is the point where a flow stream occupies the minimum area after passing through a restriction. The location and area of the vena contracta depends on the geometry of the restriction andthe characteristics of the flow.

Viscosity - The measurement of resistance of a fluid to change shape. Thecentipoise (cp) isthe mostcommon unit of measurement. Fluids with a high viscosity are "thicker" than fluids with a low viscosity.

Volumetric Meter - Measures volumetric flow rate directly, without any compensation for fluid property changes.

Weight / Time Calibration - The preferred method of calibrating a differential producer. This method uses a precisely calibrated timer and weight scale integrated into a flow loop with control valves, a diverter and a read-out instrument. Taking the amount of flow, over time, and comparing it to the readings from the device being calibrated produce the meter coefficient. The accuracy of the calibration is a function of the quality of the lab. Generally, calibrated accuracies of ± 0.25% to ± 0.5% are achieved.

Range, Span, Elevation and Suppression*

Range - The region between the limits within which a quantity is measured, received, or transmitted, expressed by stating the Lower and Upper Range Values.

Lower Range Limits - The range of input values over which a device can be zeroed. (e.g., 3 psi output)

Lower Range Value - The actual value of input within the Lower Range Limits to which a device has been zeroed.

Upper Range Limits - The range of input values over which a device can be adjusted to obtain its maximum output. (e.g., 15 psi output)

Upper Range Value - The actual value of input within the Upper Range Limits to which a device has been adjusted to obtain its maximum output.

Span - The algebraic difference between the Upper and Lower Range Values.

Span Limits - A range of input spans over which a device can be adjusted.

Table 39. Span and Range Values

Typical Range	Name	Range	Lower Range Value	Upper Range Value	Span
0 +100	zero based	0 to 100	0	+100	100
20 +100	suppressed zero range	20 to 100	20	+100	80
-25 0 +100	elevated zero range	-25 to +100	-25	+100	125
-100 0	elevated zero range	-100 to 0	-100	0	100
-100 -20	elevated zero range	-100 to -20	-100	-20	80

* Range, Span, Elevation and Supression terms from SAMASTD PMC 20.1 – 1973

Table 40. Conversion Factors: Pressure

PSI	ATM	Kg / cm²	in. H_2O (68° F)	mm Hg (32° F)	in. Hg (32° F)	kPa	BAR
1	0.068046	0.070307	27.7276	51.7150	2.03602	6.895	0.06895
14.696	1	1.0332	407.484	760	29.921	101.325	1.01325
14.2233	0.96784	1	394.38	735.559	28.959	98.096	0.98067
0.036092	0.002454	0.00253	1	1.8651	0.073430	0.249	0.00249
0.0193368	0.0013158	0.0013595	0.53616	1	0.03937	0.1333	0.001333
0.491154	0.033421	0.03453	13.6185	25.4000	1	3.3864	0.033864
0.1450	0.00987	0.010197	4.0186	7.5006	0.2953	1	0.01
14.5038	0.98692	1.01972	402.156	750.062	29.53	100	1

Table 41. Conversion Factors: Volume

Gallon (U.S.)	Cubic Feet	Cubic Inches	Barrels (Oil)	Cubic CM	Cubic Meter	Liter	Imp Gallon
1	0.1337	231	0.02381	3785	0.003785	3.785	0.8327
7.481	1	1728	0.1781	28320	0.028320	28.32	6.229
0.004329	0.0005787	1	0.0001031	16.39	0.00001639	0.01639	0.003605
42	5.615	9702	1	159000	0.159	158.94	34.97
0.000264	0.0000353	0.06102	6.29×10^{-6}	1	10^{-6}	0.001	0.000220
264.17	35.3144	61023	6.2906	10^{6}	1	1000	220.1
0.264	0.0353	61.03	0.0629	1000	0.001000	1	0.220
1.201	0.1606	277.4	0.02860	4546	0.004546	4.546	1

Table 42. Conversion Factors: Mass

lb.	oz.	Kg	gm	gal. H_2O (32°F)	Ton	Metric Tonne
1	16	0.4536	453.6	0.1198	0.0004464	0.0004536
0.0625	1	0.02835	28.35	0.00747	2.79×1^{-5}	2.84×10^{-5}
2.205	35.27	1	1000	0.2642	9.839×10^{-4}	10^{-3}
0.002205	0.03527	0.001	1	0.0002642	9.839×10^{-7}	10^{-6}
8.345	133.5	3.785	3785	1	0.003726	0.003786
2000	32,000	907.2	9.071×10^{6}	239.60	1	.907
2204.6	35280	1000	10^{6}	264.11	0.9842	1

Table 43. Distance

Inch	Foot	Centimeter	Meter
1	0.083	2.54	0.0254
12	1	30.48	0.3048
0.3937	0.0328	1	0.010
39.37	3.28	100	1

Table 44. Time

Seconds	Minutes	Hours	Days
1	0.0167	2.778 x 10-4	1.16×10^{-5}
60	1	0.0167	6.94×10^{-4}
3600	60	1	0.04162
86,400	1440	24	1

Table 45. Density

lb/ft^3	gm/cm^3	kg/m^3	lb/in^3
1	0.0160184	16.0184	0.0005787
62.43	1	1000	0.03613
0.06243	0.001	1	3.613×10^{-5}
1728	27.68	27679.8	1

Density - Ideal Gas

$$\rho = \left(\frac{144 \cdot MW \cdot P_f\,(psia)}{1545 \cdot T_f\,(^\circ R)} \right)$$

Volumetric to Mass Flow (Ideal Gas)

$$W\,(lb/hr) = Q\,(scfh) \cdot MW \cdot 0.002634$$

Temperature Conversions

The bold columns of figures refer to temperature, either Celsius or Fahrenheit. In converting from Fahrenheit to Celsius the equivalent temperature is found in the left-hand column; while converting from Celsius to Fahrenheit, the equivalent temperature is found in the right-hand column- Example: 100°F is 37.8°C; 100°C is 212°F.

Table 49. Temperature Conversions

C		F	C		F	C		F
-4.0	**-40**	-40.0	-15.0	**5**	41.0	-1.1	**30**	86.0
-36.9	**-38**	-36.4	-14.4	**6**	42.8	-0.6	**31**	87.8
-37.8	**-36**	-32.8	-13.9	**7**	44.6	0.0	**32**	89.6
-36.7	**-34**	-29.2	-13.3	**8**	46.4	0.6	**33**	91.4
-35.6	**-32**	-25.6	-12.8	**9**	48.2	1.1	**34**	93.2
-34.4	**-30**	-22.0	-12.2	**10**	50.0	1.7	**35**	95.0
-33.3	**-28**	-18.4	-11.7	**11**	51.8	2.2	**36**	96.8
-32.2	**-26**	-14.8	-11.1	**12**	53.6	2.8	**37**	98.6
-31.1	**-24**	-11.2	-10.6	**13**	55.4	3.3	**38**	100.4
-30.0	**-22**	-7.6	-10	**14**	57.2	3.9	**39**	102.2
-28.9	**-20**	-4.0	-9.4	**15**	59.0	4.4	**40**	104.0
-27.8	**-18**	-0.4	-8.9	**16**	60.8	5.0	**41**	105.8
-26.7	**-16**	3.2	-8.3	**17**	62.6	5.6	**42**	107.6
-25.6	**-14**	6.8	-7.8	**18**	64.4	6.1	**43**	109.4
-24.4	**-12**	10.4	-7.2	**19**	66.2	6.7	**44**	111.2
-23.3	**-10**	14.0	-6.7	**20**	68.0	7.2	**45**	113.0
-22.2	**-8**	17.6	-6.1	**21**	69.8	7.8	**46**	114.8
-21.1	**-6**	21.2	-5.6	**22**	71.6	8.3	**47**	116.6
-20.0	**-4**	24.9	-5.0	**23**	73.4	8.9	**48**	118.4
-18.9	**-2**	28.4	-4.0	**24**	75.2	9.4	**49**	120.2
-17.8	**0**	32.0	-3.9	**25**	77.0	10.0	**50**	122.0
-17.2	**1**	33.8	-3.3	**26**	78.8	10.6	**51**	123.8
-16.7	**2**	35.6	-2.8	**27**	80.6	11.1	**52**	125.6
-16.1	**3**	37.4	-2.2	**28**	82.4	11.7	**53**	127.4
-15.6	**4**	39.2	-1.7	**29**	84.2	12.2	**54**	129.2

Table 50. Specific Gravity of Selected Liquids

Liquid		Temp °C	Sp Gr
Acetic Acid		20/4	1.050
Acetone		15.6/15.6	.792
Alcohol, ethyl		20/4	.789
Alcohol, methyl		20/4	.792
Aluminum chloride	10%	20/4	1.073
	20%	20/4	1.154
	40%	20/4	1.342
Ammonium hydrodide	10%	15/4	.960
	20%	15/4	.925
	30%	15/4	.895
Benzene		20/4	.879
Chlorine		-33.6/4	1.560
Ethylene glycol	50%	15.6/15.6	1.070
Gasoline		15.6/15-6	.751
Glycerin		0/4	1.260
Hydrochloric acid	10%	20/4	1.047
	20%	20/4	1.098
	30%	20/4	1.149
Kerosene		15.6/15.6	.820
Mercury		15.6/15.6	13.550
Nitric acid	10%	20/4	1.0554
	30%	20/4	1.180
	50%	20/4	1.310
Oil – Linseed		15/4	.942
Phosphoric acid		20/4	1.870
Potassium hydroxid	10%	15/4	1.092
	30%	15/4	1.291
	50%	15./4	1.514
Silicone (DC-200)		15.6/15.6	.920
Sodium chloride	10%	20/4	1.071
	20%	20/4	1.48
Sodium hydroxide	10%	20/4	1.109
	30%	20/4	1.328
	50%	20/4	1.525
Sulfur		572/4	1.78
Sulfuric acid	10%	20/4	1.066
	20%	20/4	1.139
	30%	20/4	1.218
	40%	20/4	1.303
Turpentine		15.6/15.6	.873
Water		15.6/15.6	1.000
Water – Sea		15/4	1.025

Table 51. Density and Specific Gravity of Selected Gases

Gas or Vapor	Formula	Density* lbs/ft³	Sp Gr Air = 1
Acetylene	C_2H_2	0.07323	0.9073
Air	–	.08071	1.0000
Ammonia	NH_3	.04813	0.596
Argon	A	.11135	1.3796
Butane (n)	C_4H_{10}	.15725	2.0854
Butane, iso-	C_4H_{10}	.1669	2.067
Carbon dioxide	CO_2	.12341	1.5290
Carbon monoxide	CO	.07806	0.9671
Carbon oxtsulfide	COS	.170	2.10
Chlorine	Cl_2	.2006	2.486
Chlorine monoxide	Cl_2O	.243	3.01
Ethane	C_2H_6	.08469	1.0493
Ethylene	C_2H_4	.07868	0.9749
Fluorine	F_2	.1059	1.312
Helium	He	.01114	0.13804
Hydrogen	H_2	.00561	0.06952
Hydrogen bromide	HBr	.2275	2.8189
Hydrogen chloride	HCl	.10233	1.2678
Hydrogen iodide	HI	.3614	4.4776
Hydrogen sulfide	H_2S	.09608	1.190
Krypton	Kr	.2315	2.868
Methane	CH_4	.04475	0.5544
Methylamine	CH_3NH_2	.08715	1.080
Methyl chloride	CH_3Cl	.1441	1.7848
Methyl ether	$(CH_3)_2O$	.1317	1.6318
Methyl fluoride	CH_3F	.09646	1.1951
Neon	Ne	.05621	0.69638
Nitric oxide	NO	.08367	1.0366
Nitrogen	N_2	.07807	0.96724
Nitrogen (atm.)	–	.07846	0.9721
Nitrous oxide	N_2O	.1235	1.5297
Nitroxyl fluoride	NO_2F	.181	2.24
Oxygen	O_2	.08921	1.10527
Ozone	O_3	.1338	1.658
Phosphine	PH_3	.09548	1.1829
Phosphorous oxyfluoride	POF_3	.30	3.7
Phosphorous pentafluoride	PF_5	.363	4.494
Propane	C_3H_8	.1254	1.554
Sulfur dioxide	SO_2	.1827	2.2638

* Density at 32°F and 1 atmosphere

Table 52. Properties of Saturated Steam

Pressure PSIA	Temp °F	Specific Volume cu ft lb	Pressure PSIA	Temp °F	Specific Volume cu ft lb
0.20	53.14	1526.0	10.0	193.21	38.42
0.25	59.30	1235.3	11.0	197.75	35.14
0.30	64.47	1039.5	12.0	201.96	32.40
0.35	68.93	898.5	13.0	205.88	30.06
0.40	72.86	791.9	14.0	209.56	28.04
0.45	76.38	708.5	14.696	212.0	26.80
0.50	79.58	641.4	15.0	213.03	26.29
0.60	85.21	540.0	16.0	216.32	24.75
0.70	90.08	466.9	17.0	219.44	23.39
0.80	94.38	411.7	18.0	222.41	22.17
0.90	98.24	368.4	19.0	225.24	21.08
1.0	101.74	333.6	20.0	227.96	20.089
1.2	107.92	280.9	21.0	230.57	19.192
1.4	113.26	243.0	22.0	233.07	18.375
1.6	117.99	214.3	23.0	235.49	17.627
1.8	122.23	191.8	24.0	237.82	16.938
2.0	126.08	173.73	25.0	240.07	16.303
2.2	129.62	158.85	26.0	242.25	15.715
2.4	132.89	146.38	27.0	244.36	15.710
2.6	135.94	135.78	28.0	246.41	14.663
2.8	138.79	126.65	29.0	248.40	14.189
3.0	141.48	118.71	30.0	250.33	13.746
3.5	147.57	102.72	31.0	252.22	13.330
4.0	152.97	90.63	32.0	254.05	12.940
4.5	157.83	81.16	33.0	255.84	12.572
5.0	162.24	73.52	34.0	257.58	12.226
5.5	166.30	67.24	35.0	259.28	11.898
6.0	170.66	61.98	36.0	260.95	11.588
6.5	173.56	57.50	37.0	262.57	11.294
7.0	176.56	53.64	38.0	264.16	11.015
7.5	179.94	50.29	39.0	265.72	10.750
8.0	182.86	47.34	40.0	267.25	10.498
8.5	185.64	44.73	41.0	268.74	10.258
9.0	188.28	42.40	42.0	270.21	10.029
9.5	190.80	40.31	43.0	271.64	9.810

Table 52. Properties of Saturated Steam (Cont'd)

Pressure PSIA	Temp °F	Specific Volume cu ft lb	Pressure PSIA	Temp °F	Specific Volume cu ft lb
44.0	273.05	9.601	79.0	311.16	5.537
45.0	274.44	9.401	80.0	312.03	5.472
46.0	275.80	9.209	81.0	312.89	5.408
47.0	277.13	9.025	82.0	313.74	5.346
48.0	278.45	8.848	83.0	314.59	5.285
49.0	279.74	8.678	84.0	315.42	5.226
50.0	281.01	8.515	85.0	316.25	5.168
51.0	282.26	8.359	86.0	317.07	5.111
52.0	283.49	8.208	87.0	317.88	5.055
53.0	284.70	8.062	88.0	318.68	5.001
54.0	285.90	7.922	89.0	319.48	4.948
55.0	287.07	7.787	90.0	320.27	4.896
56.0	288.23	7.656	91.0	321.06	4.845
57.0	289.07	7.529	92.0	321.83	4.796
58.0	290.50	7.407	93.0	322.60	4.747
59.0	291.61	7.289	94.0	323.36	4.699
60.0	292.71	7.175	95.0	324.12	4.652
61.0	293.79	7.064	96.0	324.87	4.606
62.0	294.85	6.957	97.0	325.61	4.561
63.0	295.90	6.853	98.0	326.35	4.517
64.0	296.94	6.752	99.0	327.08	4.474
65.0	297.97	6.655	100.0	327.81	4.432
66.0	298.99	6.560	101.0	328.53	4.391
67.0	299.99	6.468	102.0	329.25	4.350
68.0	300.98	6.378	103.0	329.96	4.310
69.0	301.96	6.291	104.0	330.66	4.271
70.0	302.92	6.206	105.0	331.36	4.232
71.0	303.88	6.124	106.0	332.05	4.194
72.0	304.83	6.044	107.0	332.74	4.157
73.0	305.76	5.966	108.0	333.42	4.120
74.0	306.68	5.890	109.0	334.10	4.084
75.0	307.60	5.816	110.0	334.77	4.049
76.0	308.50	5.743	111.0	335.44	4.015
77.0	309.40	5.673	112.0	336.11	3.981
78.0	310.29	5.604			

Table 52. Properties of Saturated Steam (Cont'd)

Pressure PSIA	Temp °F	Specific Volume cu ft lb	Pressure PSIA	Temp °F	Specific Volume cu ft lb
113.0	336.77	3.947	143.0	354.67	3.155
114.0	337.42	3.914	144.0	355.21	3.134
115.0	338.07	3.882	145.0	355.76	3.114
116.0	338.72	3.850	146.0	356.29	3.094
117.0	339.36	3.819	147.0	356.83	3.074
118.0	339.99	3.788	148.0	357.36	3.054
119.0	340.62	3.758	149.0	357.89	3.034
120.0	341.25	3.728	150.0	358.42	3.015
121.0	341.88	3.699	152.0	359.46	2.977
122.0	342.50	3.670	154.0	360.49	2.940
123.0	343.11	3.642	156.0	361.52	2.904
124.0	343.72	3.614	158.0	362.53	2.869
125.0	344.33	3.587	160.0	363.53	2.834
126.0	344.94	3.560	162.0	364.53	2.801
127.0	345.54	3.533	164.0	365.51	2.768
128.0	346.13	3.507	166.0	366.48	2.736
129.0	346.73	3.481	168.0	367.45	2.705
130.0	345.32	3.455	170.0	368.41	2.675
131.0	347.90	3.430	172.0	369.35	2.645
132.0	347.90	3.430	174.0	370.29	2.616
133.0	349.06	3.381	176.0	371.22	2.587
134.0	349.64	3.357	178.0	372.14	2.559
135.0	350.21	3.333	180.0	373.06	2.532
136.0	350.78	3.310	182.0	373.96	2.505
137.0	351.35	3.287	184.0	374.86	2.479
138.0	351.91	3.264			
139.0	352.47	3.242			
140.0	353.02	3.220			
141.0	353.57	3.198			
142.0	354.12	3.177			

A: Less than .005" per Year
B: Less than .020" per Year
C: Greater than .020" per Year
S: Generally Suitable
X: Not Recommended
-: No Data

*Table 53. Corrosion Rates**

	Carbon Steel	316 SST	Carpenter 20	Hastelloy C	Monel	Nickel	Titanium	Tantalum	Platinum	Teflon	Kynar	Urethane	Natural Rubber	Neoprene
Acetic Acid (Room Temp)	C	A	A	A	B	C	A	A	A	S	S	X	X	X
Acetic Anhydride	B	B	B	A	B	B	A	A	A	-	X	X	X	S
Acetone	B	A	A	A	A	A	A	A	A	-	X	X	S	S
Alcohols	B	A	A	A	A	A	A	A	A	S	S	X	S	S
Aluminum Chloride (10-80%)	C	C	C	C	C	C	A	A	A	S	S	S	S	S
Aluminum Potassium Sulfate (Alum)	C	B	B	B	B	B	A	A	A	S	S	S	S	S
Anhydrous Ammonia	B	A	A	B	B	B	B	B	A	S	S	S	S	S
Ammonium Hydroxide (Room Temp)	B	A	A	A	C	C	A	B	A	S	S	S	S	S
Benzene	B	B	A	B	B	B	A	A	A	S	S	X	S	X
Calcium Chloride	B	C	B	A	B	B	A	A	A	S	S	S	S	S
Calcium Hydroxide	C	B	B	A	A	B	A	A	A	S	S	S	S	S
Calcium Hypochlorite	C	-	B	C	B	C	A	A	A	S	S	S	S	X
Chlorine Gas (Dry)	B	A	B	A	A	A	C	A	-	S	S	X	-	S
Chlorine Gas (Wet)	C	C	C	A	C	C	C	A	-	S	S	X	S	X
Chromic Acid	C	C	C	A	C	C	A	A	A	S	S	X	X	X

* Based on information from manufacturers' data. Since corrosion involves many more variables than this table considers, such as trace contaminants, aeration or temperature-concentration profile, the table should be used only as a guide in narrowing the choice of materials that merit further investigation. Suitability of a particular material is best determined by field test.

Table 53. Corrosion Rates (Cont'd)

	Carbon Steel	316 SST	Carpenter 20	Hastelloy C	Monel	Nickel	Titanium	Tantalum	Platinum	Teflon	Kynar	Urethane	Natural Rubber	Neoprene
Citric Acid	C	B	B	B	B	B	A	A	A	S	S	S	S	S
Copper Sulfate	C	B	B	A	B	C	A	A	A	S	S	S	S	S
Ferric Chloride	C	C	C	B	C	C	A	A	I	S	S	S	S	X
Ferrous Sulfate	C	B	B	B	C	C	A	A	I	S	S	X	S	I
Fruit Juice	C	C	A	C	A	B	A	A	A	S	S	I	X	X
Glucose (50%)	I	A	A	A	A	I	A	A	A	S	S	S	S	S
Hydrochloric Acid	C	C	C	B	C	C	A	A	C	S	S	X	S	X
Hydrogen Chloride Gas (Dry)	C	A	A	A	I	I	I	A	I	I	S	X	S	I
Hydrofluric Acid	C	C	C	B	B	B	C	C	A	I	S	X	S	X
Hydrogen Peroxide	C	B	B	A	I	B	B	A	I	S	S	I	X	X
Lactic Acid	C	B	B	A	B	C	A	A	A	S	S	S	S	S
Nitric Acid (Room Temp)	C	A	A	C	C	C	A	A	I	S	S	X	X	X
Phenol	B	B	B	B	B	B	A	A	I	S	S	X	S	S
Phosphoric Acid	C	B	B	A	C	C	B	A	I	S	S	S	S	S
Potassium Chloride	C	C	C	A	B	B	A	A	I	S	S	S	S	S
Potassium Hydroxdide	B	B	B	B	A	A	C	C	I	S	S	S	S	S
Sea Water	C	A	A	A	A	B	A	A	I	S	S	S	S	S
Sodium Carbonate	B	B	A	I	B	B	A	B	I	S	S	S	S	S
Sodium Chloride	C	B	B	A	B	B	A	A	I	S	S	S	S	S
Sodium Hydroxide (Room Temp)	B	B	B	B	A	A	A	C	A	S	S	S	S	S

Table 53. Corrosion Rates (Cont'd)

	Carbon Steel	316 SST	Carpenter 20	Hastelloy C	Monel	Nickel	Titanium	Tantalum	Platinum	Teflon	Kynar	Urethane	Natural Rubber	Neoprene
Sodium Nitrate (to 50%)	B	A	A	B	B	B	A	A	A	S	S	S	S	S
Sodium Sulfate	B	B	B	B	B	B	A	A	A	S	S	S	S	S
Sodium Sulfite	I	B	B	A	C	I	A	A	A	S	I	S	S	S
Steam	A	A	A	A	A	A	A	A	A	X	S	X	I	X
Sulfur Dioxide (Wet Gas)	C	B	B	A	C	C	C	I	I	S	S	X	I	A
Sulfuric Acid (Room Temp)	C	C	A	A	C	C	C	A	A	S	S	A	S	A
Sulfurous Acid	C	C	B	B	C	C	A	A	I	S	S	I	I	X

Table 54. Velocity in Ft/Sec of H_2O Thru 0.5" – 4" Schedule 40 Pipe

GPM	0.5" Pipe	1" Pipe	1.5" Pipe	2" Pipe	3" Pipe	4" Pipe
1.0	1.06					
1.2	1.27					
1.4	1.48					
1.6	1.69					
1.8	1.90					
2.0	2.11					
2.2	2.32					
2.4	2.53					
2.6	2.75					
2.8						
3.0	3.17	1.11				
4	4.22	1.49				
5	5.28	1.86				
6	6.33	2.23	0.95			
7						
8	8.45	2.97	1.26			
9						
10	10.56	3.71	1.58	0.96		
15	15.84	5.57	2.37	1.43		
20	21.12	7.43	3.16	1.91		
25	26.40	9.28	3.94	2.39	1.09	
30	31.68	11.14	4.73	2.87	1.30	
35		12.99	5.52	3.35	1.52	
40		14.85	6.30	3.83	1.74	1.01
45		16.70	7.09	4.30	1.95	1.13
50		18.56	7.88	4.78	2.17	1.26
60		22.27	9.47	5.74	2.60	1.51
70		25.98	11.05	6.70	3.04	1.76

Table 54. Vel. in Ft/Sec H_2O Thru 0.5" – 4" Schedule 40 Pipe (Cont'd)

GPM	0.5" Pipe	1" Pipe	1.5" Pipe	2" Pipe	3" Pipe	4" Pipe
80		29.70	12.62	7.65	3.47	2.02
90		33.4	14.20	8.60	3.91	2.27
100			15.76	9.56	4.34	2.52
125			19.70	11.97	5.43	3.15
150			23.64	14.36	6.51	3.78
175			27.58	16.75	7.60	4.41
200			31.52	19.12	8.68	5.04
225				21.51	9.77	5.67
250				23.90	10.85	6.30
275				26.29	11.94	6.93
300				28.68	13.00	7.56
325				31.07	14.12	8.19
350					15.19	8.82
375					16.28	9.45
400					17.36	10.08
425					18.45	10.71
450					19.53	11.34
475					20.62	11.97
500					21.70	12.60
550					23.87	13.85
600					26.04	15.12
650					28.21	16.38
700					30.38	17.64

Table 55. Velocity in Ft/Sec of H_2O Thru 6"– 16" Schedule 40 Pipe

GPM	6" Pipe	8" Pipe	10" Pipe	12" Pipe	14" Pipe	16" Pipe
60						
70						
80						
90	0.90					
100	1.11					
125	1.39					
150	1.67	0.96				
175	1.94	1.12				
200	2.22	1.28				
225	2.50	1.44				
250	2.78	1.60	1.02			
275	3.05	1.76	1.12			
300	3.33	1.92	1.22			
325	3.61	2.08	1.32			
350	3.89	2.24	1.42	1.00		
375	4.16	2.40	1.52	1.07		
400	4.44	2.56	1.62	1.14		
425	4.72	2.73	1.72	1.22	1.01	
450	5.00	2.89	1.83	1.28	1.06	
475	5.27	3.04	1.93	1.36	1.13	
500	5.55	3.21	2.03	1.43	1.19	
550	6.11	3.53	2.24	1.58	1.30	
600	6.66	3.85	2.44	1.72	1.42	1.09
650	7.22	4.17	2.64	1.76	1.54	1.18
700	7.78	4.49	2.85	2.01	1.66	1.27
750	8.33	4.81	3.05	2.15	1.77	1.36
800	8.88	5.13	3.25	2.29	1.89	1.45
850	9.44	5.45	3.46	2.44	2.01	1.54

Table 55. Vel. in Ft/Sec H_2O Thru 6" – 16" Schedule 40 Pipe (Cont'd)

GPM	6" Pipe	8" Pipe	10" Pipe	12" Pipe	14" Pipe	16" Pipe
900	9.99	5.77	3.66	2.58	2.13	1.63
950	10.55	6.09	3.86	2.72	2.25	1.73
1000	11.10	6.41	4.07	2.87	2.37	1.82
1100	12.22	7.05	4.48	3.15	2.61	1.99
1200	13.33	7.70	4.88	3.44	2.85	2.18
1300	14.43	8.33	5.29	3.73	3.08	2.36
1400	15.55	8.98	5.70	4.01	3.32	2.54
1500	16.66	9.62	6.10	4.30	3.56	2.72
1600	17.77	10.26	6.51	4.59	3.79	2.91
1700	18.88	10.90	6.92	4.87	4.03	3.09
1800	19.99	11.54	7.32	5.16	4.27	3.27
1900	21.11	12.18	7.73	5.45	4.51	3.45
2000	22.22	12.82	8.14	5.73	4.74	3.63
2500	27.78	16.03	10.17	7.17	5.93	4.54
3000	33.33	19.24	12.20	8.60	7.2	5.45
3500		22.44	14.24	10.03	8.30	6.36
4000		25.65	16.27	11.47	9.49	7.26
4500		28.87	18.31	12.90	10.67	8.17
5000		32.06	20.35	14.33	11.86	9.08
6000			24.41	17.20	14.23	10.90
7000			28.49	20.07	16.60	12.71
8000				22.93	18.98	14.52

Table 56. Velocity in Ft/Sec of H_2O Thru 18" – 24" Schedule 40 Pipe

GPM	18" Pipe	20" Pipe	24" Pipe
1500			
1600			
1700			
1800	2.58		
2000	2.88		
2500	3.59		
3000	4.31	3.45	
3500	5.03	4.03	
4000	5.74	4.61	3.19
4500	6.47	5.19	3.59
5000	7.17	5.76	3.99
6000	8.62	6.92	4.80
7000	10.04	8.06	5.68
8000	11.48	9.23	6.38
9000	12.92	10.37	7.19
10000	14.37	11.53	7.96
12000	17.23	13.83	9.57
14000	20.10	16.14	11.18
16000	22.96	18.43	12.77

Table 57. Physical Properties of Water

Temp. °F	Saturation Pressure PSIA	Density lb/ft³	Specific Gravity	Weight lb/gal.
32	.08854	62.42	1.001	8.345
40	.12170	62.42	1.001	8.345
50	.17811	62.38	1.000	8.340
60	.2563	62.34	1.000	8.334
70	.3631	62.27	.998	8.325
80	.5069	62.19	.997	8.314
90	.6982	62.11	.996	8.303
100	.9492	62.00	.994	8.289
110	1.2748	61.84	.991	8.267
120	1.6924	71.73	.990	8.253
130	2.2225	61.54	.987	8.227
140	2.8886	61.39	.984	8.207
150	3.718	61.20	.981	8.182
160	4.741	61.01	.978	8.156
170	5.992	60.79	.975	8.127
180	7.510	60.57	.971	8.098
190	9.339	60.35	.968	8.068
200	11.526	60.13	.964	8.039
210	14.123	59.88	.960	8.005
212	14.696	59.81	.959	7.996
220	17.186	59.63	.956	7.972
240	24.969	59.10	.947	7.901
260	35.429	58.51	.938	7.822
280	49.203	57.94	.929	7.746
300	67.013	57.31	.919	7.662
350	134.63	55.59	.891	7.432
400	247.31	53.65	.860	7.172
450	422.6	51.55	.826	6.892
500	680.8	49.02	.786	6.553
550	1045.2	45.87	.735	6.132
600	1542.9	42.37	.679	5.664
700	3093.7	27.10	.434	3.623

Table 58. Pipe Diameters

Nominal Pipe Diameter	Inside Pipe Diameter										
	Carbon Steel Pipe						Stainless Steel Pipe			Iron Pipe	
	Sched 10	Sched 20	Sched 40	Sched 60	Sched 80	Sched 160	Sched 10S	Sched 40S	Sched 80S	ST	EX
1/2	–	–	0.622	–	0.546	0.466	0.6674	0.622	0.546	0.622	0.546
3/4	–	–	0.824	–	0.742	0.612	0.884	0.824	0.742	0.824	0.742
1	–	–	1.049	–	0.957	0.815	1.097	1.049	0.957	1.049	0.957
1-1/4	–	–	1.380	–	1.278	1.160	1.442	1.380	1.278	1.350	1.278
1-1/2	–	–	1.610	–	1.500	1.338	1.682	1.610	1.500	1.610	1.500
2	–	–	2.067	–	1.939	1.687	2.157	2.067	1.939	2.067	1.939
2-1/2	–	–	2.469	–	2.323	2.125	2.635	2.469	2.323	2.469	2.323
3	–	–	3.068	–	2.900	2.624	3.260	3.068	2.900	3.068	2.900
4	–	–	4.026	–	3.826	3.438	4.260	4.026	3.826	4.026	3.826
5	–	–	5.047	–	4.813	4.313	5.295	5.047	4.813	5.047	4.813
6	–	–	6.065	–	5.761	5.187	6.357	6.065	5.761	6.065	5.761
8	–	8.125	7.981	7.813	7.625	6.813	8.329	7.981	7.625	7.981	7.625
10	–	10.250	10.020	9.750	9.562	8.500	10.420	10.020	9.750	10.020	9.750

Table 58. Pipe Diameters (Cont'd)

Nominal Pipe Diameter	Inside Pipe Diameter										
	Carbon Steel Pipe						Stainless Steel Pipe			Iron Pipe	
	Sched 10	Sched 20	Sched 40	Sched 60	Sched 80	Sched 120	Sched 10S	Sched 40S	Sched 80S	ST	EX
12	–	12.250	11.938	11.626	11.374	10.126	12.390	12.000	11.750	12.00	11.750
14	13.500	13.376	13.124	12.812	12.500	11.188	13.624	–	–	13.250	13.000
16	15.500	15.376	15.000	14.688	14.312	12.812	15.624	–	–	15.250	15.000
18	17.500	17.376	16.876	16.500	16.124	14.438	17.624	–	–	17.250	17.000
20	19.500	19.250	18.812	18.376	17.938	16.062	19.564	–	–	19.250	19.000
22	21.500	21.250	–	20.250	19.750	17.750	21.564	–	–	21.250	21.000
24	23.500	23.250	22.624	22.062	21.562	19.312	23.500	–	–	23.250	23.000
26	25.376	25.000	–	–	–	–	–	–	–	25.250	25.000
28	27.376	27.000	–	–	–	–	–	–	–	27.250	27.000
30	29.376	29.000	–	–	–	–	29.376	–	–	29.250	29.000
32	31.376	31.000	–	–	–	–	–	–	–	31.250	31.000
34	33.312	33.000	32.624	–	–	–	–	–	–	33.250	33.000
36	35.376	35.000	34.500	–	–	–	–	–	–	35.250	35.000

Table 59. Orifice Coefficient (K) Values – 2" Pipe

Beta	Reynolds #							
	8,000	10,000	15,000	20,000	30,000	50,000	100,000	500,000
0.100	0.6048	0.6046	0.6043	0.6041	0.6039	0.6037	0.6036	0.6035
.150	.6016	.6011	.6005	.6002	.5999	.5997	.5995	.5983
.200	.6007	.6000	.5991	.5986	.5982	.5979	.5976	.5974
.250	.6021	.6012	.6001	.5995	.5990	.5985	.5982	.5979
.300	.6058	.6047	.6032	.6025	.6018	.6012	.6007	.6005
.350	.6110	.6099	.6089	.6070	.6061	.6053	.6049	.6043
.400	.6185	.6166	.6142	.6129	.6117	.6107	.6099	.6095
.450	.6284	.6259	.6225	.6209	.6192	.6179	.6169	.6161
.500	.6419	.6384	.6339	.6316	.6293	.6275	.6261	.6250
.550	.6600	.6553	.6490	.6458	.6436	.6402	.6384	.6368
.600	.6837	.6773	.6688	.6644	.6601	.6567	.6541	.6521
.625	.6984	.6910	.6809	.6759	.6708	.6668	.6638	.6614
.650	.7150	.7062	.6945	.6886	.6828	.6781	.6746	.6718
.675	.7338	.7238	.7095	.7028	.6962	.6908	.6868	.6836
.700	.7546	.7429	.7272	.7192	.7115	.7053	.7006	.6968

Table 60. Orifice Coefficient (K) Values – 3" Pipe

Beta	Reynolds #							
	8,000	10,000	15,000	20,000	30,000	50,000	100,000	500,000
0.100	0.6004	0.6001	0.5996	0.5992	0.5992	0.5990	0.5989	0.5988
.150	.5983	.5975	.5967	.5963	.5961	.5958	.5955	.5953
.200	.5992	.5984	.5974	.5968	.5963	.5958	.5955	.5952
.250	.6027	.6017	.6002	.5994	.5988	.5982	.5978	.5974
.300	.6067	.6055	.6037	.6028	.6019	.6012	.6007	.6002
.350	.6123	.6106	.6085	.6073	.6061	.6052	.6045	.6039
.400	.6201	.6178	.6147	.6132	.6117	.6104	.6094	.6089
.450	.6306	.6275	.6233	.6213	.6192	.6175	.6163	.6153
.500	.6444	.6408	.6351	.6322	.6294	.6271	.6254	.6241
.550	.6651	.6591	.6510	.6471	.6427	.6398	.6369	.6354
.600	.6917	.6834	.6722	.6666	.6610	.6566	.6532	.6506
.625	.7080	.6982	.6850	.6785	.6719	.6667	.6628	.6596
.650	.7266	.7151	.6998	.6922	.6845	.6782	.6737	.6700
.675	.7477	.7343	.7163	.7973	.6983	.6911	.6858	.6815
.700	.7715	.7559	.7350	.7245	.7143	.7058	.6996	.6946

Data and Conversion Tables

Table 61. Orifice Coefficient (K) Values – 4" Pipe

Beta	Reynolds #							
	8,000	10,000	15,000	20,000	30,000	50,000	100,000	500,000
0.100	0.5975	0.5971	0.5967	0.5964	0.5961	0.5959	0.5958	0.5957
.150	.5974	.5066	.5956	.5951	.5947	.5943	.5941	.5940
.200	.6000	.5990	.5977	.5970	.5963	.5958	.5955	.5951
.250	.6038	.6025	.6006	.5995	.5989	.5984	.5979	.5975
.300	.6081	.6065	.6044	.6034	.6023	.6015	.6008	.6003
.350	.6135	.6116	.6090	.6076	1.6064	.6053	.6045	.6039
.400	.6214	.6188	.6152	.6135	.6119	.6105	.6094	.6086
.450	.6324	.6289	.6242	.6218	.6194	.6175	.6161	.6150
.500	.6481	.6431	.6364	.6330	.6296	.6271	.6251	.6235
.550	.6692	.6625	.6535	.6484	.6439	.6400	.6371	.6350
.600	.6991	.6891	.6757	.6690	.6623	.6569	.6530	.6498
.625	.7173	.7054	.6896	.6817	.6737	.6674	.6626	.6588
.650	.7378	.7238	.7054	.6958	.6864	.6790	.6735	.6690
.675	.7613	.7448	.7229	.7120	.7010	.6923	.6857	.6804
.700	.7878	.7685	.7421	.7301	.7172	.7071	.6994	.6932
.725	.8182	.7958	.7760	.7512	.7362	.7253	.7163	.7081
.750	.8557	.8295	.7946	.7774	.7600	.7461	.7356	.7272

Table 62. Orifice Coefficient (K) Values – 6" Pipe

Beta	Reynolds #							
	8,000	10,000	15,000	20,000	30,000	50,000	100,000	500,000
0.100	0.5956	0.5950	0.5942	0.5938	0.5934	0.5931	0.5929	0.5927
150	.5984	.5974	.5961	.5955	.5949	.5942	.5938	.5936
.200	.6021	.6008	.5990	.5981	.5972	.5965	.5960	.5955
.250	.6058	.6042	.6020	.6009	.5998	.5990	.5982	.5977
.300	.6100	.6081	.6055	.6041	.6028	.6018	.6012	.6003
.350	.6159	.6135	.6103	.6087	.6071	.6058	.6048	.6040
.400	.6241	.6209	.6166	.6145	.6124	.6107	.6095	.6085
.450	.6366	.6322	.6263	.6232	.6202	.6179	.6161	.6148
.500		.6481	.6396	.6354	.6311	.6276	.6250	.6231
.550		.6708	.6583	.6521	.6459	.6410	.6372	.6343
.600		.7014	.6836	.6747	.6658	.6587	.6533	.6490
.625		.7207	.6993	.6887	.6781	.6695	.6632	.6580
.650		.7425	.7171	.7044	.6918	.6817	.6741	.6681
.675		.7673	.7374	.7214	.7974	.6954	.6857	.6795
.700		.7952	.7600	.7425	.7254	.7109	.7003	.6919
.725		.8274	.7862	.7666	.7451	.7306	.7163	.7067
.750		.8658	.8179	.7937	.7698	.7505	.7361	.7246

Table 63. Orifice Coefficient (K) Values – 8" Pipe

Beta	Reynolds #							
	8,000	10,000	15,000	20,000	30,000	50,000	100,000	500,000
0.100	0.5956	0.5949	0.5940	0.5935	0.5930	0.5926	0.5923	0.5921
.150	.5998	.5986	.5969	.5961	.5952	.5946	.5942	.5938
.200	.6039	.6022	.6000	.5989	.5977	.5969	.5863	.5958
.250	.6075	.6056	.6029	.6016	.6003	.5992	.5984	.5978
.300	.6119	.6096	.6065	.6049	.6034	.6022	.6013	.6005
.350	.6179	.6151	.6112	.6093	.6073	.6059	.6047	.6038
.400	.6266	.6229	.6180	.6155	.6130	.6110	.6095	.6084
.450	.6400	.6348	.6278	.6243	.6209	.6181	.6161	.6141
.500	.6603	.6527	.6425	.6373	.6325	.6284	.6252	.6229
.550	.6892	.6780	.6631	.6556	.6481	.6422	.6376	.6341
.600	.7296	.7131	.6912	.6803	.6691	.6605	.6539	.6486
.625	.7551	.7353	.7088	.6956	.6824	.6719	.6640	.6575
.650	.7839	.7603	.7237	.7129	.6971	.6847	.6751	.6676
.675	.8169	.7889	.7515	.7329	.7142	.6992	.6880	.6790
.700	.8542	.8212	.7772	.7552	.7332	.7154	.7023	.6918
.725		.8586	.8067	.7808	.7548	.7340	.7185	.7062
.750		.9022	.8416	.8111	.7809	.7565	.7383	.7272

Table 64. Orifice Coefficient (K) Values – 10" Pipe

Beta	Reynolds #							
	8,000	10,000	15,000	20,000	30,000	50,000	100,000	500,000
0.100	0.5962	0.5953	0.5942	0.5936	0.5930	0.5926	0.5922	0.5919
.150	.6005	.5995	.5975	.5968	.5958	.5950	.5944	.5939
.200	.6054	.6035	.6009	.5996	.5983	.5972	.5965	.5959
.250	.6092	.6069	.6042	.6027	.6012	.6000	.5988	.5980
.300	.6136	.6110	.6074	.6056	.6039	.6025	.6015	.6007
.350	.6196	.6164	.6129	.6100	.6079	.6063	.6050	.6040
.400	.6287	.6246	.6191	.6163	.6147	.6114	.6097	.6084
.450	.6433	.6374	.6296	.6257	.6218	.6187	.6163	.6144
.500	.6656	.6570	.6454	.6396	.6337	.6291	.6257	.6229
.550	.6989	.6857	.6681	.6593	.6505	.6435	.6381	.6339
.600	.7441	.7247	.6988	.6859	.6728	.6626	.6548	.6486
.625	.7720	.7487	.7176	.7021	.6865	.6742	.6649	.6575
.650	.8059	.7778	.7404	.7216	.7028	.6878	.6766	.6676
.675	.8439	.8104	.7656	.7432	.7209	.7031	.6895	.6788
.700		.8474	.7943	.7678	.7413	.7201	.7042	.6915
.725		.8897	.8272	.7960	.7647	.7397	.7210	.7060
.750		.9370	.8644	.8735	.7753	.7408	.7299	.7213

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Table 68. Conductivities of Aqueous Solutions

Chemical Name	Formula	% By Weight	Temp °F (°C)	Conductivity µS/cm
Acetic Acid	CH_3CO_2H	10	64 (18)	1.53×10^3
Ammonia	NH_3	4.01	59 (15)	1.10×10^3
Butryic Acid	$C_3H_7CO_2H$	15.03	64 (18)	9.55×10^2
Cadmium Chloride	C_dCl_2	5	64 (18)	1.67×10^4
Cadmium Sulfate	C_dSO_4	5	64 (18)	1.46×10^4
Calcium Chloride	C_aCl_2	25	64 (18)	1.78×10^5
Cupric Sulphate	$CuSO_4$	10	64 (18)	3.20×10^4
Formic Acid	HCO_2H	39.95	64 (18)	9.84×10^3
Hydrobromic Acid	HBr	10	59 (15)	3.55×10^5
Hydrochloric Acid	HCl	20	59 (15)	7.62×10^5
Hydroflouric Acid	HF	1.5	64 (18)	1.98×10^4
Lithium Chloride	LiCl	10	64 (18)	1.22×10^5
Lithium Sulfate	$LiSO_4$	10	59 (15)	6.10×10^4
Magnesium Chloride	$MgCl_2$	20	64 (18)	1.40×10^5
Magnesium Sulfate	$MgSO_4$	10	59 (15)	4.14×10^4
Nitric Acid	HNO_3	31	64 (18)	7.82×10^5
Phosphoric Acid	H_3PO_4	35	59 (15)	1.86×10^5
Potassium Bromide	KBr	20	59 (15)	1.91×10^5
Potassium Chloride	KCl	15	64 (18)	2.02×10^5
Potassium Hydroxide	KOH	16.8	59 (15)	4.46×10^5
Potassium Nitrate	KNO_3	15	64 (18)	1.19×10^5
Potassium Sulfate	K_2SO_4	10	64 (18)	8.6×10^4
Potassium Sulfide	K_2S	9.93	64 (18)	2.34×10^5
Proprionic Acid	$C_3H_6O_2$	20.02	64 (18)	1.04×10^3

Table 68. Conductivities of Aqueous Solutions (Cont'd)

Chemical Name	Formula	% By Weight	Temp (°C)	Conductivity µS/cm
Sodium Acetate	$C_2H_3N_3O_2$	20	64 (18)	6.51×10^4
Sodium Carbonate	NA_2CO_3	10	64 (18)	7.05×10^4
Sodium Chloride	NACl	15	64 (18)	1.64×10^5
Sodium Hydroxide	NAOH	20	64 (18)	2.72×10^5
Sodium Sulfate	NA_2SO_4	10	64 (18)	6.87×104
Sulfuric Acid	H_2SO_4	50	64 (18)	5.41×10^5
Zinc Chloride	$ZnCl_2$	20	59 (15)	9.12×10^4
Zinc Sulfate	$ZnSO_4$	15	64 (18)	4.15×10^4

Table 69. Viscosity of Various Gases

Temp °F	Viscosity Centipoise
Acetylene	
323	.00935
Air	
32	.0171
104	.0190
444	.0264
633	.0305
674	.0312
768	.0341
Ammonia	
32	.0092
68	.0098
212	.0128
302	.0146
482	.0181
Argon	
32	.0210
68	.0222
212	.0269
392	.0322
Benzene	
0	.0065
40	.0070
70	.0075
100	.0080
200	.0091
Butane	
0	.0075
40	.0080
70	.0085
100	.0090
200	.0104
Butylene	
66	.0074
212	.0095
Carbon Dioxide	
144	.0090
76	.0106
32	.0139
68	.0148
86	.0153
Carbon Monoxide	
32	.0166
59	.0172
260.8	.0218
Chlorine	
122	.0147
212	.0168
302	.0187
392	.0208
Ethane	
32	.0085
63	.0090
Ethylene	
32	.0091
68	.0101
122	.0110
212	.0126
Helium	
32	.0186
68	.0194
Hydrogen	
-172	.0057
-143.5	.0062
-25	.0077
32	.0084
69	.0088
264	.0182
Hydrogen Chloride	
54.4	.0139
61.8	.0141
212	.0182
Hydrogen Sulfide	
32	.0117
62.6	.0124
212	.0159
Methane	
32	.0102
68	.0109
212	.0133
Nitrogen	
-6.7	.0156
51.6	.0171
81	.0178
261	.0219
440	.0256
Oxygen	
32	.0189
67	.0202
262	.0257
440	.0302

Table 69. Viscosity of Various Gases (Cont'd)

Temp °F	Viscosity Centipoise
n-Pentane	
77	.0068
212	.0101
Propane	
64.2	.0079
213	.0101
Propylene	
62	.0083
122	.0093
Sulfur Dioxide	
32	.0116
64.4	.0124
68.9	.0125
213	.0161

Table 70. Viscosity of Various Liquids

Temp °F	Viscosity Centipoise
Acetic Acid	
59	1.31
64	1.30
77	1.155
86	1.04
106	1.00
212	.43
Acetic Anhydride	
32	1.24
59	.971
64	.90
86	.783
212	.490
Acetone	
14.0	.450
32.0	.399
59.0	.337
77.0	.316
Ammonia	
-92	.475
-58	.317
-40	.276
-28	.255
Benzene	
32	.912
50	.758
68	.652
86	.564
104	.503
122	.442
Carbon Tetrachloride	
32	1.329
59	1.038
68	.969
86	.843
104	.739
140	.585
Chlorine Liquid	
-40	.505
-20	.462
20	.400
60	.350
100	.313
Ethlbenzene	
63	.691
Ethylene Glycol	
68	19.9
104	9.13
140	4.95
176	3.02
Ethylene Oxide	
-57	.577
37	.488
5.8	.394
32.0	.320

Table 70. Viscosity of Various Liquids (Cont'd)

Temp °F	Viscosity Centipoise
Fluorobenzine	
68	.598
140	.389
212	.275
Glycerine	
32	12110
43	6260
59.0	2330
68	1490
77	954
86	629
Heptane	
32	.524
63	.461
68	.409
77	.386
104	.271
Hexane	
32	.401
68	.326
77	.294
104	.271
Hydrochloric Acid 31.5%	
0	3.4
20	2.9
40	2.5
60	2.0
80	1.8
100	1.6
140	.2

Temp °F	Viscosity Centipoise
Iodine Liquid	
241	2.27
Isoheptane	
32	.481
68	.384
104	.315
Isohexane	
32	.376
68	.306
104	.254
Isopentane	
32	.273
68	.223
Methyl Alcohol (Methane)	
-48	1.98
32	.82
59	.623
68	.597
86	.510
Methyl Chloride	
0	.25
20	.23
40	.21
60	.19
100	.16
Naphthalene	
176	.967
212	.776

Temp °F	Viscosity Centipoise
Nitric Acid	
32	2.275
Nitrobenzene	
37	2.91
42	2.71
50	2.48
68	2.03
Nitromethane	
32	.853
77	.620
n-Octane	
32	.706
68	.524
104	.433
Pentane	
32	.289
68	.240
Phenol	
65	2.7
122	3.49
158	2.03
194	1.26
Phosphorous Liquid	
71	2.34
88	2.01
110	1.73
123	1.60
140	1.45
158	1.32

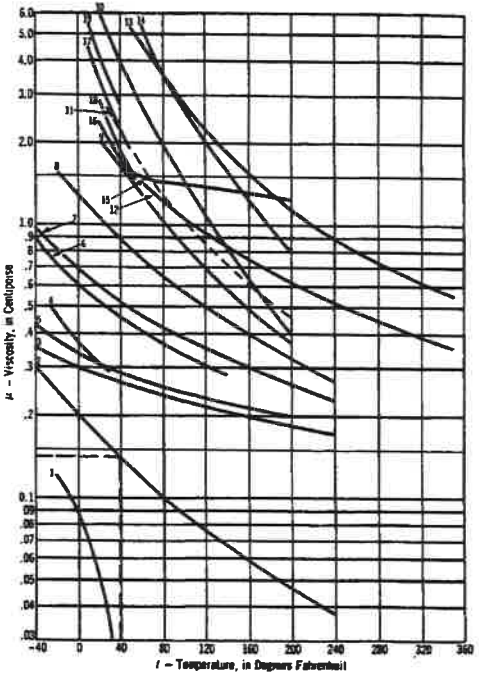
Table 70. Viscosity of Various Liquids (Cont'd)

Temp °F	Viscosity Centipoise
Sodium Hydroxide	
70	100
100	40
120	25
140	15
160	9.5
200	3.7
220	2.4
2.50	1.4
Sodium Liquid	
0	2.4
26	1.3
40	1.2
60	1.1
100	1.0
140	.85

Temp °F	Viscosity Centipoise
Sulfur (Gas Free)	
253	10.9
276	8.7
301	77.1
314	7.2
317	7.6
319	14.5
Sulfur Dioxide	
-28	.5508
13	.4285
32	.3936

Temp °F	Viscosity Centipoise
Sulfuric Acid	
32	48.4
59	32.8
68	25.4
86	15.7
104	11.5
122	8.82
140	7.22

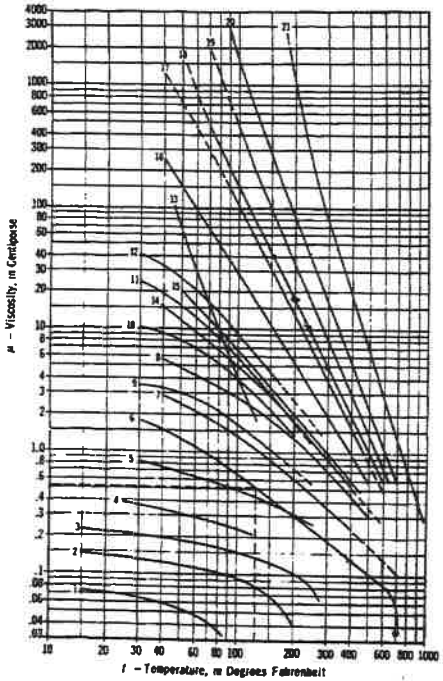
Table 71. Viscosity Curves for Selected Liquids



1. Carbon Dioxide	CO_2	11. 20% Sulphuric Acid	20% H_2SO_4
2. Ammonia	NH_3	12. Dowtherm E	
3. Methyl Chloride	CH_3Cl	13. Dowtherm A	
4. Sulphur Dioxide	SO_2	14. 20% Sodium Hydroxide	20% NaOH
5. Freon 12	F-12	15. Mercury	
6. Freon 114	F-114	16. 10% Sodium Chloride Brine	10% NaCl
7. Freon 11	F-11	17. 20% Sodium Chloride Brine	20% NaCl
8. Freon 113	F-113	18. 10% Calcium Chloride Brine	20% $CaCl_2$
9. Ethyl Alcohol		19. 20% Calcium Chloride Brine	20% $CaCl_2$
10. Isopropyl Alcohol			

Example: The viscosity of ammonia at 40°F is ° 14 centipoise.

Table 72. Viscosity Curves for Selected Liquid Petroleum Products



1. Ethane ($C_2 H_6$)
2. Propane ($C_3 H_8$)
3. Butane ($C_4 H_{10}$)
4. Natural Gasoline
5. Gasoline
6. Water
7. Kerosene
8. Distillate
9. 48 Deg. API Crude
10. 40 Deg. API Crude
11. 35.6 Deg. API Crude
12. 32.6 Deg. API Crude
13. Salt Creek Crude
14. Fuel 3 (Max.)
15. Fuel 5 (Min.)
16. SAE 10 Lube (100 V.I.)
17. SAE 30 Lube (100 V.I.)
18. Fuel 5 (max.) or Fuel 6 (Min.)
19. SAE 70 Lube (100 V.I.)
20. Bunker C Fuel (Max.) & M. C. Residuum
21. Asphalt

Example: The viscosity of water at 125°F is °52 centipiose (Curve 6)

Table 73. Viscosity of Water

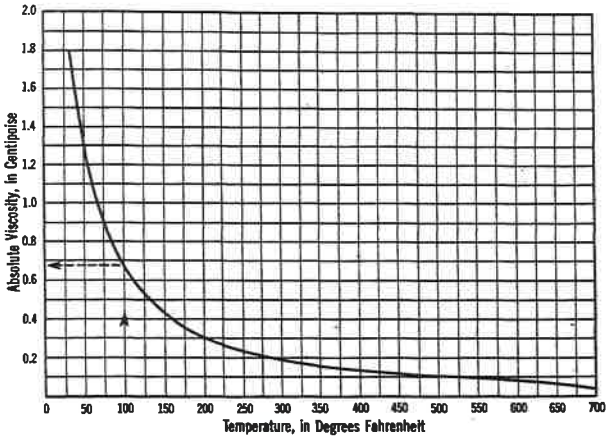


Table 74. Orifice Expansion Factor (F_a)

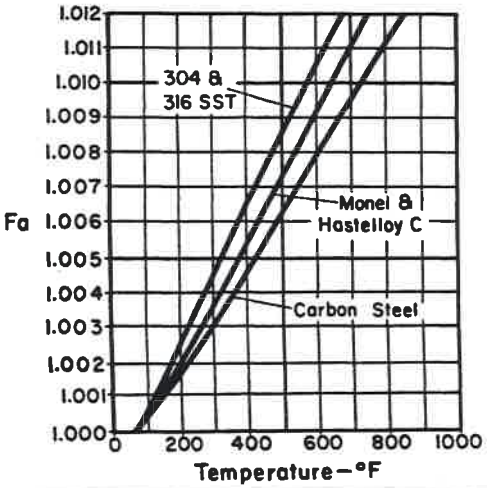
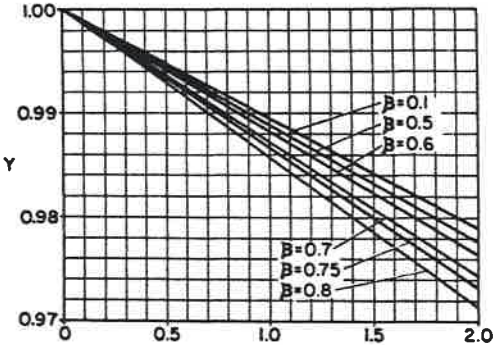


Table 75. Gas Expansion Factor (Y)



For H/D	use	β
0.2		0.38
0.3		0.50
0.4		0.60
0.5		0.70

Table 76. Correction Factors for Specific Gravity, Pressure and Temperature for Flowing Conditions

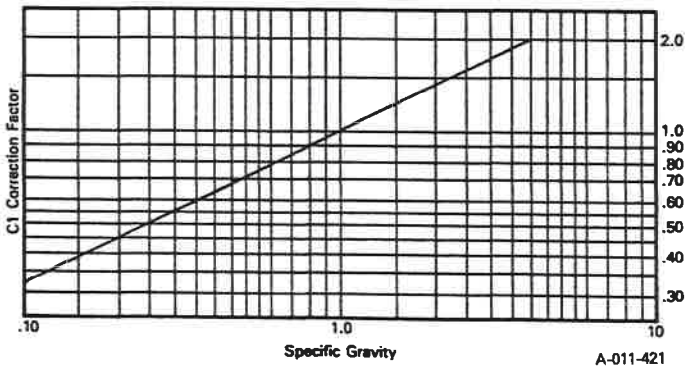


Table 77. Correction Factors for Specific Gravity, Pressure and Temperature for Flowing Conditions

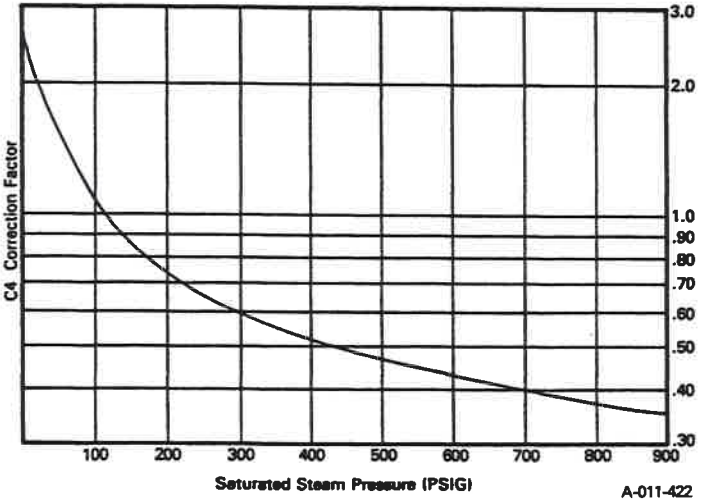


Table 78. Correction Factors for Specific Gravity, Pressure and Temperature for Flowing Conditions

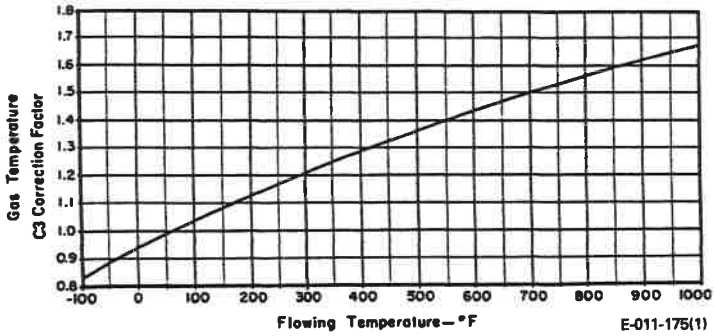


Table 79. Correction Factors for Specific Gravity, Pressure and Temperature for Flowing Conditions

NOTES

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